



Derivatives for climate positioning

by

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Disclaimer

I, Syed Aasiful Islam, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

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Abstract

This research based dissertation investigates the design of climate-related financial instruments, with a particular focus on weather risk in agricultural commodity markets. Climate change has increased uncertainty surrounding agricultural production, creating demand for financial products that enable market participants to hedge or express views on climate-related risks. Coffee is selected as the primary case study because of its strong sensitivity to weather conditions and climate variability. Historical market and climate data are analysed to identify the key drivers of Arabica and Robusta coffee prices and to assess whether physical climate variables contain predictive information for future price movements. These findings are then used to develop quantitative investment strategies that provide exposure to climate-related risks through traded coffee futures. This approach is considered as a practical alternative to constructing bespoke climate derivatives, which face challenges arising from the non-tradeability of climate variables, pricing, replication, and hedging. The proposed strategies are evaluated through historical back-testing, after which their practical implementation and commercial viability as investable products are considered.

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Chapter 1

Introduction

Climate change has emerged as one of the most significant sources of long-term financial risk, influencing the valuation of assets, investment decisions, and the stability of financial markets. Unlike traditional macroeconomic risks, climate-related risks are characterised by deep uncertainty regarding both their timing and magnitude. Generally speaking climate risk can be split into three interconnected categories: physical risk, transition risk, and liability risk (Little 2015 bank of England) [35].

Physical climate risk refers to the economic and financial losses arising from climate-related hazards. These hazards include both acute events, such as floods, storms and wildfires, and chronic changes, including rising temperatures, changing precipitation patterns and prolonged drought. The severity of physical climate risk depends not only on the hazard itself but also on the exposure and vulnerability of affected assets, sectors and economies. [40]

Transition risk refers to the financial risks arising from the adjustment towards a low-carbon economy. These risks emerge from changes in climate policy, regulation, technology, market preferences, consumption preferences and changes in legal frameworks. The introduction of carbon taxes, emissions trading schemes and environmental regulation can significantly alter a firms expected future cash flows, particularly for carbon-intensive industries creating significant financial risk for both investors and institutions [40]

Liability risk is the risk that an entity is held legally or financially responsible for climate-related damages, whether through litigation, regulatory enforcement, or shareholder action [41]. It is the slowest-moving of the three chains, and the universe of directly affected firms is comparatively narrow. Liability risk is included here for completeness and conceptual contrast with the other two channels; for the reasons of liquidity and long time horizon, it is not the primary focus of the derivative design proposed in this dissertation, though it is a candidate for future market development.

Due to these risks investors, policymakers and firms frequently hold differing expectations regarding the pace of the global transition towards a greener economy. These heterogeneous beliefs create opportunities for financial markets to aggregate information and facilitate risk sharing, while simultaneously generating demand for financial instruments that allow

investors to express differing views on future climate outcomes. Our objective with this project is to therefore create products that allow expression of these beliefs whilst also allowing investors to hedge against a price change caused by climate.

1.1 Existing related products

Financial markets already offer a number of products designed to manage or provide exposure to climate-related risks. Traditional weather derivatives, such as Heating Degree Day (HDD) and Cooling Degree Day (CDD) contracts, provide payouts based on deviations of temperature from predetermined thresholds and are primarily used by energy companies to hedge weather-related demand risk. Environmental derivatives, including futures on the European Union Emissions Trading System (EU ETS), allow firms and financial market participants to manage exposure to carbon prices and the transition risks associated with climate policy. In addition, prediction market platforms such as Kalshi and Polymarket offer contracts linked to the occurrence of climate-related events, enabling participants to express views on future climate outcomes. Climate-related insurance products, including drought and natural disaster insurance, also provide protection against adverse climate events by transferring specific risks to insurers. Despite the availability of these instruments, each addresses only a particular aspect of climate risk and has limitations in terms of market scope, liquidity, accessibility or the range of risks that can be expressed. Consequently, there remains a lack of liquid and widely accessible financial instruments that enable investors and producers to express broader views on climate change or hedge against its combined financial impacts, motivating the development of new climate-positioning instruments.

1.2 What this project did

This dissertation focuses on the design of a QIS for the coffee commodity market. Although coffee futures are among the most actively traded agricultural contracts, they primarily allow market participants to hedge against overall price fluctuations arising from a combination of macroeconomic conditions, supply and demand dynamics, policy changes and climate effects. They do not, however, provide direct protection against climate-specific risks. Coffee production is particularly sensitive to physical climate factors such as temperature, rainfall and drought, with adverse weather events affecting both crop yields and quality. As a result, producers remain exposed to significant climate-related risks, while consumers ultimately bear the consequences through increased price volatility. This dissertation therefore investigates whether climate-related information can be used to construct a quantitative investment strategy that provides more targeted exposure, or protection against, climate risk. Chapter 2 reviews the literature on climate risk, commodity markets, and asset pricing, Chapter 3 develops the mathematical background while Chapter 4 develops a working definition of a climate position. Chapter 5 analyses Arabica and Robusta coffee futures to determine whether a measurable climate signal can be extracted. Chapter 6 discusses the drawbacks and differences in creating a climate derivative as opposed to a strategy. Chapter 7 presents the strategies, followed by their back-tests. Chapter 8 concludes the dissertation.

Chapter 2

Literature Review

2.1 Climate finance literature

Although climate finance is a relatively new field, it has grown rapidly as climate change has become an increasingly important economic and financial concern. Stroebe and Wurgler (2021) [52] provide a comprehensive survey of finance academics, industry professionals, regulators, and policy economists to assess current views on climate finance. The authors find that respondents regard transition risks, particularly those arising from future climate regulation, as the most significant source of financial risk over the next five years, while physical climate risks are expected to dominate over longer investment horizons. An overwhelming majority of respondents also believe that financial markets currently underestimate climate risks, implying that climate-related information is not yet fully incorporated into asset prices. Overall, the study highlights a broad consensus that climate change will become increasingly important for financial markets and that understanding and pricing climate risk remains a key area for future research.

One of the other foundational papers in this area is Climate Finance by Giglio, Kelly, and Stroebe (2021) [21]. The authors show that financial markets increasingly incorporate climate risk into asset prices, with firms exposed to transition risk often facing higher costs of capital, while assets subject to greater physical climate risks may trade at lower valuations. However, they also conclude that whether climate risk is fully priced remains an open question, owing to uncertainty surrounding future climate policies and the long-term nature of climate change.

Another interesting paper which also links to the previous report is the paper by Engle et al (2020) [15]. In this paper Engle et al. (2020) develop a market-based measure of climate risk by constructing a Climate Change News (CCN) index from newspaper articles in the wall street journal using textual analysis. The authors then examine how financial assets respond to innovations in this index using a linear factor model,

$$R_{i,t} = \alpha_i + \beta_i CCN_t + \varepsilon_{i,t}, \quad (2.1)$$

where $R_{i,t}$ is the return on asset i , CCN_t represents unexpected changes in climate news,

and β_i measures the asset’s exposure to climate-related risk. Stocks with larger climate betas exhibit greater sensitivity to climate news, allowing the authors to construct hedge portfolios that track climate risk. Although the study focuses on equity markets, it demonstrates that climate information can be quantified into a measurable financial risk factor. This provides a useful foundation for incorporating climate-related variables into financial models, supporting the use of weather and climate indices in the pricing and design of weather derivatives for agricultural commodities such as coffee. Other researchers have used the work of Engle in order to analyse the impact of climate news on other financial assets. Building on this work, Huynh and Xia (2021) [30] investigate whether climate change risk is also reflected in corporate bond markets. Using a climate change news index, the authors estimate each bond’s climate beta, measuring the covariance between bond returns and climate-related news. They find that bonds with higher climate betas earn lower expected returns, suggesting that investors are willing to accept lower returns for assets that provide protection against climate risk. Furthermore, bonds issued by firms with stronger environmental performance command higher prices during periods of heightened climate concern, indicating that investors place a premium on environmentally sustainable firms. These findings provide further evidence that climate-related risks are increasingly incorporated into financial asset prices and reinforce the importance of considering climate factors in financial risk management.

Other people have done similar things in order to investigate if climate risks are reflected in prices. Faccini et al (2023) [17] investigate if climate risk is reflected in stock prices. The authors employ Latent Dirichlet Allocation (LDA), an unsupervised machine learning algorithm that identifies latent topics within a collection of 13 million Reuters documents. Under LDA, each document is modelled as a mixture of K topics, while each topic is represented by a probability distribution over words. Formally,

$$\theta_d \sim \text{Dir}(\alpha), \quad \phi_k \sim \text{Dir}(\beta), \quad (2.2)$$

where θ_d denotes the topic proportions for document d , and ϕ_k is the word distribution for topic k . The algorithm identifies a climate-related topic and aggregates its intensity over time to construct a climate news index. Faccini goes into detail about creating this news index as part of the paper “Dissecting Climate Risks: Are they reflected in stock prices” (2023) [17]. The authors find that, although financial markets respond to a range of climate-related news, only transition risks associated with climate policy are consistently priced by investors. Firms with greater exposure to climate policy news are found to earn higher expected returns, suggesting that investors demand a premium for bearing transition risk. In contrast, there is limited evidence that physical climate risks, such as natural disasters and global warming, are systematically reflected in equity prices. These findings indicate that financial markets currently place greater emphasis on the uncertainty surrounding future climate policies than on the direct economic impacts of climate change, highlighting the importance of distinguishing between physical and transition climate risks in asset pricing.

In the past people have tried to create derivatives that serve the purpose of hedging against

climate risk. Bloch, Annan and Bowles (2010) [5] propose an innovative framework for managing flood and coastal risks by introducing climate derivatives linked to sea level rise. Rather than focusing solely on predicting future climate change, the paper explores how financial markets can be used to transfer climate-related risks between parties. The authors argue that governments, coastal developers and financial institutions could use derivative contracts to hedge against uncertain future sea level rise, thereby reducing the financial burden associated with climate adaptation.

The paper models climate uncertainty using stochastic processes rather than deterministic forecasts. The authors note that climate variability occurs across multiple timescales and cannot be adequately represented by a single deterministic trend. Consequently, they extend previous climate models by incorporating uncertainty in both climate sensitivity and response time, while also allowing for abrupt changes through jump-diffusion processes. This enables the model to capture both gradual warming and sudden climate shifts that may significantly affect sea level projections. A central mathematical concept is the modelling of climate risk as a first-passage problem, where a climate event occurs once sea level exceeds a predefined threshold. If τ denotes the first time the sea level crosses a predefined barrier, the value of a climate default bond depends on the probability that this event has not occurred before maturity. The bond price is given by

$$CB(t_0, T) = B(t_0, T) (1 - R P(\tau \leq T)), \quad (2.3)$$

where $B(t_0, T)$ is the risk-free bond price, R is the recovery rate, and $P(\tau \leq T)$ is the probability that the sea level exceeds the specified threshold before maturity. They also introduce a Climate Default Swap (CDS), analogous to a credit default swap, in which the buyer pays periodic premiums to a protection seller in exchange for compensation if sea level rises above a specified barrier during the contract period. The fair premium is determined using the probability distribution of the first-passage time of sea level crossing the threshold.

Despite its originality, the study has several limitations. First, this paper does not properly consider how to hedge the derivative. They argue that natural hedges exist by having two counter parties with differing views with each other transfer risk to each other but this is not a perfect solution because counter parties have different characteristics and due to the lack of a tradable underlying delta hedging is not possible and so there will always be some residual risk that cannot be hedged. Second, the model assumes that climate variables can be represented using relatively simple stochastic processes, which in reality may not fully capture the complexity of coupled climate systems. Third, the proposed market depends on the existence of sufficient buyers, sellers and reliable climate indices, yet no mature market for long-term climate derivatives currently exists. Finally, the paper largely demonstrates theoretical feasibility without validating the framework using real-world market data or analysing issues such as liquidity, transaction costs and regulatory constraints.

On another note an interesting paper which has aided the direction of this dissertation by Ma, Zhou, and Li (2024) [37] utilises the same news index by Faccini [17] in order to analyse

a basket of agricultural commodities. Using data from 2005 to 2021, the authors find that climate risk possesses significant in-sample and out-of-sample predictive power for future agricultural commodity returns, even after controlling for conventional macroeconomic and financial predictors. Among the four climate risk measures, physical climate risk particularly news related to global warming provides substantially stronger predictive power than transition risk. The study also demonstrates that incorporating climate risk into forecasting models generates meaningful economic gains for investors and suggests that climate risk influences agricultural commodity prices through its effects on supply conditions and investor attention. These findings highlight the importance of climate information in agricultural commodity pricing and motivate some of the work for the following chapters in this dissertation.

2.2 commodities and signal generation literature

Commodities markets will prove to be very useful within this dissertation due to the fact that climate significantly impacts the price of commodities. Climate events, such as droughts and frosts, directly impact coffee production. This shifts the supply curve to the left while demand remains relatively unchanged. As a result, the market reaches a new equilibrium with a higher price and a lower quantity of coffee. Therefore, climate change causes coffee prices to increase through its effect on supply. In general, the primary mechanism to trade commodities is through futures contracts which are good as they allow traders to protect themselves against future price hikes. Commodity futures also allow traders to replicate the returns of trading physical commodities without all the hassle of owning them.

The commodities market is a very important market and has been for centuries, and can be split into two categories, hard vs soft commodities. Hard commodities are natural resources that have to be mined or pumped from the ground like oil and metals. Soft commodities are things like agricultural products and livestock [50]. Unlike the equities market commodities are tangible and so the price of contracts related to commodities are heavily linked to the production, storage, and consumption of the underlying.

Storage is a very important thing in commodities. A very well known and cited paper by Holbrook Working (1949) [69] introduces the Theory of Storage, providing one of the earliest economic explanations for the relationship between spot and futures prices in commodity markets. Rather than viewing futures prices solely as reflecting expectations of future spot prices, Working argued that the spread between spot and futures prices is determined by the costs and benefits of holding physical inventories. In particular, when inventories are abundant, the benefits of holding the physical commodity are low, resulting in futures prices exceeding spot prices. Conversely, when inventories are scarce, holding inventories provides valuable insurance against supply disruptions, increasing the implicit convenience yield and causing spot prices to exceed futures prices. This inventory based explanation forms the theoretical foundation of modern commodity pricing and explains why market conditions such as contango and backwardation arise. Figure 2.1 shows the current forward curves for coffee and gold from LSEG workspace which exhibit backwardation and contango re-

spectively. Gorton Rouwenhorst (2004) [22] conduct an empirical study on the historical performance of commodities futures. They find that Contrary to the common perception that commodity futures are highly speculative investments, commodity futures have historically generated returns and Sharpe ratios comparable to those of equities, while exhibiting negative correlations with both stocks and bonds. Furthermore, commodity futures are shown to be positively correlated with inflation and unexpected inflation, making them an effective hedge against inflationary shocks. The study concludes that commodity futures provide substantial diversification benefits within a traditional investment portfolio and should be regarded as a distinct asset class rather than simply a speculative investment.



Figure 2.1: Backwardation, top, and contango, bottom.

To quantify the effect of climate on coffee prices, it is necessary to construct signals that capture relevant climatic events. Dias et al. (2024) [12] develop a range of climate-based indicators to assess the impacts of climate risks, vulnerabilities, diseases and pests on Brazilian Arabica coffee production under both current and future climate scenarios. These indicators are designed to capture specific environmental conditions that influence coffee yields. Examples are presented in Table 2.1. These include Iw, which measures water availability; TIB, which captures the frequency of temperatures outside the optimal range for Arabica coffee growth; TImax, which measures extreme heat events; TIfrost, which measures frost events; DIrust, which measures susceptibility to coffee rust; and DIbrown, which measures susceptibility to brown eye spot disease.

Although the modelling approach adopted by Dias et al. (2024) is beyond the scope of this project, their methodology provides a useful framework for constructing climate signals that capture weather-related risks. Their findings indicate that water stress is currently

Table 2.1: Arabica coffee-specific risk indicators.

Indicator	Bioclimatic range
Iw	Number of years with $P_{\text{annual}} < 1000 \text{ mm year}^{-1}$.
TIB	Number of days with $T_{\text{AS}} \leq 10.5^\circ\text{C}$ or $T_{\text{AS}} \geq 32.0^\circ\text{C}$.
TImax	Number of days with $T_{\text{ASmax}} > 34.0^\circ\text{C}$.
TIfrost	Number of days with $T_{\text{ASmin}} \leq 4.0^\circ\text{C}$.
DIrust	High risk of coffee rust: number of days with $18.0^\circ\text{C} \leq T_{\text{AS}} \leq 26.0^\circ\text{C}$ and $\text{RH} \geq 75\%$.
DIbrown	High risk of brown eye spot: number of days with $18.0^\circ\text{C} \leq T_{\text{AS}} \leq 24.0^\circ\text{C}$ and $\text{RH} \geq 75\%$.

the greatest threat to Brazilian Arabica coffee production, while future climate change is expected to increase both water and heat stress, alongside pest and disease pressures. Consequently, the authors estimate that 35–75% of existing Arabica coffee-growing areas could become economically unviable by the end of the century due to declining yields.

The question of whether weather data can be converted into a tradeable signal has a direct precedent in the frozen concentrated orange juice (FCOJ) market. Roll [44] selected FCOJ precisely because it offers an unusually clean natural experiment: over ninety per cent of the oranges used in American FCOJ production are grown in a small region of central Florida, so a single weather station captures the overwhelming majority of the supply risk facing the contract. This geographic concentration removes much of the aggregation problem that afflicts weather analysis in more dispersed crops, and it is the same property that motivates the focus on Minas Gerais in this dissertation, since that single Brazilian state accounts for roughly half of national Arabica output.

Roll’s central result is twofold, and the two halves point in opposite directions. On the one hand, he documents a statistically significant relation between FCOJ returns and subsequent errors in the temperature forecasts issued by the National Weather Service for central Florida. The direction of the relation implies that futures prices contain weather information not present in the official forecast: a higher closing price relative to the open implies a smaller anticipated crop and therefore colder weather, so the forecast could be improved by revising it downward. That prices anticipate meteorological information rather than merely responding to it is an important consideration for signal design, and it anticipates a difficulty encountered in Chapter 7, where a strategy conditioned on realised rather than forecast temperature proves to be systematically late.

On the other hand, Roll finds that although weather news helps determine futures price movements, it accounts for a relatively small share of them, with the regressions producing coefficients of determination on the order of one to two per cent. This result proved more influential than the first. It became known as the orange juice puzzle and is widely cited in the behavioural finance literature as evidence that asset prices move for reasons only loosely connected to observable fundamentals: if weather, the most obvious determinant of orange supply, explains so little of the variation in orange juice prices, the residual variation is difficult to attribute to public information at all. Roll additionally regressed returns on rainfall forecast errors and found no discernible relation, noting that the effect of rainfall on the crop is considerably less clear-cut than that of temperature.

Boudoukh et al. [7] revisit this puzzle and offer a resolution that is directly relevant to the methodology adopted here. Their argument is that the weak explanatory power in Roll's regressions is a consequence of specification rather than of market irrationality. The relationship between temperature and crop damage is not linear: oranges are essentially unaffected by temperatures above freezing and severely damaged below it, so the response function is closer to a threshold than a slope. A linear regression of returns on temperature averages this discontinuity away. When the nonlinearity is modelled explicitly, the authors find that freezing temperatures account for approximately half of the variation in FCOJ returns, a figure an order of magnitude larger than Roll's.

Taken together, these two papers establish both the precedent for the present analysis and a methodological warning. The precedent is that a geographically concentrated, freeze-exposed crop traded through a liquid futures contract is the canonical setting in which weather signals can be extracted from prices. The warning is that the functional form of the signal matters more than its statistical significance in a linear specification. This informs the design of the strategies in Chapter 7, which condition on a temperature *threshold* rather than on a continuous temperature variable, on the grounds that frost damage to Arabica is likewise a discontinuous function of minimum temperature. It also provides a context for the modest coefficients of determination reported in the predictive regressions of Chapter 5, which are consistent in magnitude with Roll's original findings and should be interpreted in that light rather than as evidence that no relationship exists.

Chapter 3

Mathematical Background

3.1 Derivatives and asset pricing literature

The asset-pricing theory required for this dissertation is standard, and is drawn principally from Shreve's Stochastic Calculus for Finance II [48] and Björk's Arbitrage Theory in Continuous Time [4]. This section proceeds in three stages: the probabilistic and stochastic-calculus foundations, the continuous-time market model built upon them, and the risk-neutral pricing theory that follows. Throughout, the apparatus is stated in its conventional form.

3.1.1 Probability Space

Uncertainty is modelled on a probability space

$$(\Omega, \mathcal{F}, P), \tag{3.1}$$

where Ω is the sample space, $\mathcal{F}$ a σ -algebra of measurable events, and P the physical probability measure $P : \mathcal{F} \rightarrow [0, 1]$ satisfying

$$P(\Omega) = 1, \quad P(\emptyset) = 0. \tag{3.2}$$

Each $\omega \in \Omega$ represents one possible future state of the world, and $\mathcal{F}$ collects the events to which probabilities may be assigned. This three provides the foundation on which random variables, stochastic processes and contingent claims are defined. In climate finance, Ω is interpreted broadly enough to contain climate-transition states, so that financial and climate uncertainty are carried on a single probability space.

3.1.2 Filtration and Adaptedness

Information does not arrive all at once, and is represented by a filtration: an increasing family of σ -algebras

$$(\mathcal{F}_t)_{t \geq 0}, \quad \mathcal{F}_s \subseteq \mathcal{F}_t, \quad 0 \leq s \leq t \leq T, \tag{3.3}$$

assumed to satisfy the usual conditions, namely right-continuity. Since $\mathcal{F}_t$ represents everything observable by time t , the inclusion above formalises the fact that information accumulates rather than decays. A stochastic process $X = (X_t)_{t \geq 0}$ is adapted to $(\mathcal{F}_t)_{t \geq 0}$ if X_t is $\mathcal{F}_t$ -measurable for every t . Adaptedness excludes the use of future information and is therefore imposed on every price, portfolio and state process considered below; without it, arbitrage-free pricing arguments have no content.

3.1.3 Martingales

An adapted process $(X_t)_{t \geq 0}$ with $\mathbb{E}|X_t| < \infty$ for all t is a martingale with respect to $(\mathcal{F}_t)_{t \geq 0}$ and P if

$$\mathbb{E}^P[X_t | \mathcal{F}_s] = X_s, \quad 0 \leq s \leq t. \quad (3.4)$$

The martingale property states that the best forecast of a future value, given current information, is the current value. Martingales are central to what follows because the absence of arbitrage will be characterised precisely as the existence of a measure under which discounted prices are martingales.

3.1.4 Brownian Motion

A standard Brownian motion $(W_t)_{t \geq 0}$ is a stochastic process satisfying

1. $W_0 = 0$;
2. $W_t - W_s \sim N(0, t - s)$ for $0 \leq s < t$;
3. increments over disjoint intervals are independent;
4. sample paths are almost surely continuous.

Brownian motion is itself a martingale with respect to its natural filtration and constitutes the fundamental source of randomness in continuous-time finance. Its paths are almost surely nowhere differentiable, which is why ordinary calculus must be replaced by the Itô calculus below.

3.1.5 Itô Processes and Itô's Lemma

An *Itô process* is a stochastic process of the form

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t, \quad (3.5)$$

equivalently

$$X_t = X_0 + \int_0^t \mu(X_s, s) ds + \int_0^t \sigma(X_s, s) dW_s, \quad (3.6)$$

where μ is the drift, σ the diffusion coefficient, and the coefficients satisfy the integrability conditions $\int_0^T |\mu(X_s, s)| ds < \infty$ and $\int_0^T \sigma(X_s, s)^2 ds < \infty$ almost surely.

For $f \in C^{1,2}$, Itô's lemma gives the dynamics of $f(t, X_t)$ as

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW_t. \quad (3.7)$$

The second-order term, absent from ordinary calculus, arises from the quadratic variation $d\langle W \rangle_t = dt$. Itô's lemma is the principal computational tool of derivative pricing and is used repeatedly.

3.1.6 Asset Price Processes and the Physical Measure

Consider a market of n risky assets with adapted price processes

$$S_t = (S_t^1, S_t^2, \dots, S_t^n), \quad (3.8)$$

whose dynamics under the physical measure P are given by the Itô system

$$dS_t^i = \mu^i(S_t, t) dt + \sigma^i(S_t, t) dW_t, \quad i = 1, \dots, n. \quad (3.9)$$

Under P , the drift μ^i is the expected rate of return actually observed in the market. The canonical specification is geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P, \quad S_t = S_0 \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t^P \right\}, \quad (3.10)$$

which guarantees $S_t > 0$ and admits a closed-form solution.

The market also contains a risk-free money-market account B , used as numéraire, with

$$dB_t = r B_t dt, \quad B_t = e^{rt}, \quad B_0 = 1, \quad (3.11)$$

for constant short rate r . Prices expressed in units of the numéraire are written

$$\tilde{S}_t^i = \frac{S_t^i}{B_t} = e^{-rt} S_t^i, \quad (3.12)$$

and it is these *discounted* prices that appear in the martingale conditions below. The framework extends to climate finance by permitting a climate-related state variable to serve as an underlying risk factor in place of, or alongside, the traded assets S .

3.1.7 Portfolio Processes and Trading Strategies

A trading strategy is an adapted process

$$\theta = (\theta_t)_{t \geq 0}, \quad \theta_t = (\theta_t^0, \theta_t^1, \dots, \theta_t^n), \quad (3.13)$$

where θ_t^i denotes the number of units of asset i held at time t and θ_t^0 the holding in the money-market account. Adaptedness of θ is the requirement that positions be chosen using only currently available information, and is what makes a strategy economically admissible.

The associated portfolio value is

$$V_t = \theta_t^0 B_t + \sum_{i=1}^n \theta_t^i S_t^i = \theta_t^\top S_t, \quad (3.14)$$

with S_t understood to include B_t as its zeroth component. Strategies may be static, with θ_t held fixed, or dynamic, with positions rebalanced continuously; it is the dynamic case that permits replication and hedging.

3.1.8 Self-Financing Portfolios

A trading strategy is *self-financing* if rebalancing requires neither the injection nor the withdrawal of external capital, so that changes in portfolio value arise solely from changes in asset prices:

$$dV_t = \theta_t^\top dS_t. \quad (3.15)$$

Equivalently, in discounted terms,

$$d\tilde{V}_t = \sum_{i=1}^n \theta_t^i d\tilde{S}_t^i, \quad \tilde{V}_t = \frac{V_t}{B_t}, \quad (3.16)$$

so that the discounted portfolio value is a stochastic integral against discounted prices alone. The self-financing condition is what makes replication arguments meaningful, since a portfolio that could be topped up externally could replicate any payoff whatsoever.

3.1.9 Arbitrage

An admissible self-financing strategy θ constitutes an *arbitrage* if its value process satisfies

$$V_0 = 0, \quad P(V_T \geq 0) = 1, \quad P(V_T > 0) > 0. \quad (3.17)$$

That is, a riskless profit is obtained from zero initial capital. A market admitting no such strategy is said to be arbitrage-free, and this condition is the economic foundation of everything that follows.

3.1.10 Contingent Claims and European Derivatives

A contingent claim maturing at T is an $\mathcal{F}_T$ -measurable random variable V_T . A *European* claim is one whose payoff depends only on the terminal value of the underlying,

$$V_T = h(S_T), \quad (3.18)$$

for a payoff function $h(\cdot)$. The standard examples are the European call and put,

$$h(S_T) = (S_T - K)^+, \quad h(S_T) = (K - S_T)^+, \quad (3.19)$$

for strike K . Path-dependent claims, by contrast, are $\mathcal{F}_T$ -measurable functionals of the entire trajectory $(S_t)_{0 \leq t \leq T}$.

3.1.11 Replication and Market Completeness

A claim V_T is attainable, or replicable, if there exists an admissible self-financing strategy θ whose terminal value reproduces it,

$$V_T = \theta_T^\top S_T, \quad V_t = \theta_t^\top S_t \text{ for all } t \in [0, T]. \quad (3.20)$$

In an arbitrage-free market the price of an attainable claim is uniquely determined: any two prices would permit a riskless profit from the difference. A market in which every contingent claim is attainable is said to be complete.

3.1.12 Equivalent Martingale Measures

A probability measure Q is equivalent to P , written $Q \sim P$, if the two agree on which events have zero probability,

$$Q(A) = 0 \iff P(A) = 0, \quad A \in \mathcal{F}. \quad (3.21)$$

Such a Q is an equivalent martingale measure, or risk-neutral measure, if every discounted price process is a Q -martingale:

$$\tilde{S}_t^i = \mathbb{E}^Q[\tilde{S}_T^i \mid \mathcal{F}_t], \quad \tilde{S}_t^i = e^{-rt} S_t^i. \quad (3.22)$$

Under Q the drift of every traded asset is replaced by the risk-free rate,

$$dS_t = rS_t dt + \sigma S_t dW_t^Q, \quad (3.23)$$

so that expected returns are equalised across assets. Equivalence guarantees that P and Q agree on which events are possible, and therefore on which strategies constitute arbitrage; the change of measure alters probabilities, not the null sets.

3.1.13 Girsanov's Theorem

Girsanov's theorem supplies the change of measure from P to Q . Define the market price of risk

$$\lambda_t = \frac{\mu_t - r}{\sigma_t}, \quad (3.24)$$

and the Radon–Nikodym density

$$\left. \frac{dQ}{dP} \right|_{\mathcal{F}_T} = \exp \left\{ - \int_0^T \lambda_s dW_s^P - \frac{1}{2} \int_0^T \lambda_s^2 ds \right\}, \quad (3.25)$$

assumed to be a true martingale, as guaranteed for instance by Novikov's condition. Then

$$W_t^Q = W_t^P + \int_0^t \lambda_s ds \quad (3.26)$$

is a standard Brownian motion under Q . The effect is to replace the physical drift μ with the risk-free rate r while leaving the diffusion coefficient σ unchanged: volatility is invariant to the change of measure, whereas expected returns are not.

3.1.14 The Fundamental Theorem of Asset Pricing

The two parts of the Fundamental Theorem of Asset Pricing tie the preceding concepts together:

1. A market is arbitrage-free if and only if there exists at least one equivalent martingale measure $Q \sim P$.
2. An arbitrage-free market is complete if and only if that equivalent martingale measure is unique.

The theorem converts the absence of arbitrage, into a probabilistic one, the existence of a measure change, and in doing so provides the foundation of modern derivative pricing.

3.1.15 Risk-Neutral Valuation

For an attainable claim with payoff V_T , combining the self-financing condition with the martingale property of discounted prices gives the valuation formula

$$V_t = B_t \mathbb{E}^Q \left[\frac{V_T}{B_T} \middle| \mathcal{F}_t \right] = \mathbb{E}^Q \left[e^{-r(T-t)} V_T \mid \mathcal{F}_t \right]. \quad (3.27)$$

The price of a derivative is therefore the discounted expected payoff taken under the risk-neutral measure rather than the physical one. Crucially, the physical drift μ does not appear as prices are determined by no-arbitrage considerations alone and are independent of investor risk preferences.

3.2 Principal Component Analysis

Principal Component Analysis (PCA) is a dimension reduction technique used to represent a high-dimensional dataset by a smaller number of uncorrelated linear combinations of the original variables. The aim is to replace a p -dimensional vector with a lower-dimensional representation while retaining as much of the variation in the data as possible. PCA is particularly useful when the original variables are correlated, since much of the information in the data may then be captured by a smaller number of common directions. [29]

Let

$$X \in \mathbb{R}^p$$

be a random vector with finite second moments. Denote its mean by

$$\mu = \mathbb{E}[X],$$

and define the centred random vector

$$Y = X - \mu.$$

The covariance matrix of X is

$$\Sigma = \mathbb{E} \left[(X - \mu)(X - \mu)^\top \right] = \mathbb{E} \left[Y Y^\top \right].$$

The matrix Σ is symmetric and positive semi-definite.

For any unit vector $v \in \mathbb{R}^p$, the scalar projection of the centred data onto the direction v is

$$Z(v) = v^\top Y.$$

This gives a one-dimensional summary of the original random vector. PCA chooses the direction v so that this one-dimensional projection is as informative as possible, where informativeness is measured by variance.

Since

$$\text{Var}(v^\top Y) = \mathbb{E} \left[(v^\top Y)^2 \right] = v^\top \Sigma v,$$

the first principal direction is defined as the solution to the constrained optimisation problem

$$v_1 = \arg \max_{\|v\|=1} v^\top \Sigma v.$$

The constraint $\|v\|=1$ is required because otherwise the variance could be made arbitrarily large simply by scaling v .

This optimisation problem leads directly to an eigenvalue problem. Consider the Lagrangian

$$\mathcal{L}(v, \lambda) = v^\top \Sigma v - \lambda(v^\top v - 1).$$

Differentiating with respect to v gives

$$2\Sigma v - 2\lambda v = 0,$$

and hence

$$\Sigma v = \lambda v.$$

Therefore, the principal directions are eigenvectors of the covariance matrix Σ , and the corresponding eigenvalues give the variances of the principal component scores.

Let the eigenvalues of Σ be ordered as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0,$$

with corresponding orthonormal eigenvectors

$$v_1, v_2, \dots, v_p.$$

The first principal component is then

$$Z_1 = v_1^\top Y,$$

and more generally, the k -th principal component is

$$Z_k = v_k^\top Y, \quad k = 1, \dots, p.$$

The first component Z_1 has the largest possible variance among all one-dimensional linear projections of Y . The second component Z_2 has the largest possible variance among all projections orthogonal to v_1 , and so on.

Since the eigenvectors are orthonormal, the principal component scores are uncorrelated. For $j \neq k$,

$$\text{Cov}(Z_j, Z_k) = \mathbb{E} \left[(v_j^\top Y)(v_k^\top Y) \right] = v_j^\top \Sigma v_k = \lambda_k v_j^\top v_k = 0.$$

Thus PCA transforms the original possibly correlated variables into a new set of uncorrelated variables.

The variance of the k -th principal component is

$$\text{Var}(Z_k) = v_k^\top \Sigma v_k = \lambda_k.$$

Hence the total variance in the original centred vector is

$$\mathbb{E} [\|X - \mu\|^2] = \sum_{k=1}^p \lambda_k.$$

The proportion of total variance explained by the first K principal components is therefore

$$\text{CPV}(K) = \frac{\sum_{k=1}^K \lambda_k}{\sum_{k=1}^p \lambda_k}.$$

This quantity is often used to decide how many principal components to keep.

Using only the first $K < p$ principal components, the random vector X can be approximated by

$$X_K = \mu + \sum_{k=1}^K Z_k v_k.$$

This is the orthogonal projection of X onto the subspace spanned by the first K principal directions. The mean squared reconstruction error is

$$\mathbb{E} [\|X - X_K\|^2] = \sum_{k=K+1}^p \lambda_k.$$

Therefore, keeping the first K principal components minimises the reconstruction error among all K -dimensional linear approximations.

3.3 Performance Measures

3.3.1 The Sharpe ratio

The Sharpe ratio is one of the most well known metrics in modern financial mathematics. It was proposed by William F. Sharpe [46] in 1966 and helps to evaluate the efficiency of an investment strategy by accounting for the adjusted risk required for that return. Prior to this practitioners focused on mainly the raw return without much of an account for the risk.

Let $r_{p,t}$ denote the portfolio return in period t , $r_{f,t}$ the corresponding risk-free rate, and T the number of observations. The sample moments are

$$\bar{R}_p = \frac{1}{T} \sum_{t=1}^T r_{p,t}, \quad \bar{R}_f = \frac{1}{T} \sum_{t=1}^T r_{f,t}, \quad \hat{\sigma}_p^2 = \frac{1}{T-1} \sum_{t=1}^T (r_{p,t} - \bar{R}_p)^2, \quad (3.28)$$

and the discrete-time Sharpe ratio is defined as

$$\text{SR} = \frac{\bar{R}_p - \bar{R}_f}{\hat{\sigma}_p}. \quad (3.29)$$

3.3.2 Sortino Ratio

The Sharpe ratio evaluates performance by dividing excess return by total volatility. This treats both positive and negative deviations from the mean as risk. However, from an investor's perspective, upside volatility is not usually undesirable, since returns above the benchmark represent favourable outcomes. The Sortino ratio modifies the Sharpe ratio by penalising only downside variation, that is, returns that fall below a benchmark or target return. Let R^ϕ denote the return of strategy ϕ , and let R^b denote the benchmark return. The Sortino ratio is defined as

$$\text{Sortino}(\phi) = \frac{\mathbb{E}[R^\phi - R^b]}{\text{sd}\left((R^\phi - R^b)^-\right)}, \quad (3.30)$$

where

$$x^- = \min(x, 0) \quad (3.31)$$

denotes the downside part of x . Therefore, $(R^\phi - R^b)^-$ captures only those observations where the strategy underperforms the benchmark. The denominator is therefore the downside deviation, rather than the full standard deviation of returns. Unless otherwise stated, the benchmark R^b is usually taken to be the risk-free return.

3.3.3 Maximum Drawdown

While volatility-based measures such as the Sharpe ratio and Sortino ratio summarise the dispersion of returns, they do not directly measure the size of sustained losses experienced by an investor. In practice, investors are often concerned with how far a strategy can fall from its previous peak, since large losses may lead to margin pressure, forced liquidation, or withdrawal of capital. Maximum drawdown captures this path-dependent risk by measuring the largest proportional decline in portfolio value over a given period.

Let S_t denote the value of an investment at time t , and consider the period $[T_0, T_1]$. The running maximum value of the investment up to time t is

$$M_t = \max_{T_0 \leq u \leq t} S_u. \quad (3.32)$$

The drawdown at time t is then defined as the percentage fall from the previous peak:

$$D_t = 1 - \frac{S_t}{M_t}. \quad (3.33)$$

The maximum drawdown over the period $[T_0, T_1]$ is the largest value of D_t :

$$\text{MDD}(S, T_0, T_1) = \max_{T_0 \leq t \leq T_1} \left(1 - \frac{S_t}{\max_{T_0 \leq u \leq t} S_u} \right). \quad (3.34)$$

Equivalently, if $\tau_{\min}$ denotes the time at which the worst drawdown is realised, then

$$\text{MDD}(S, T_0, T_1) = 1 - \frac{S_{\tau_{\min}}}{\max_{T_0 \leq u \leq \tau_{\min}} S_u}. \quad (3.35)$$

Thus maximum drawdown measures the worst peak-to-trough loss experienced by the investment. Unlike volatility, it depends on the order in which returns occur and is therefore a useful measure of the realised downside risk faced by an investor.

3.3.4 Calmar Ratio

The Calmar ratio measures performance relative to the worst peak-to-trough loss over a specified historical period. It is therefore similar in spirit to the Sharpe ratio, but replaces volatility with maximum drawdown as the measure of risk. This makes it particularly useful for strategies where large sustained losses are more important than ordinary return fluctuations. Let R^ϕ denote the return of strategy ϕ , and let S^ϕ denote the corresponding value process. The Calmar ratio is defined as

$$\text{Calmar}(\phi) = \frac{\mathbb{E}[R^\phi]}{\text{MDD}(S^\phi, -\ell, 0)}, \quad (3.36)$$

where $\text{MDD}(S^\phi, -\ell, 0)$ is the maximum drawdown of the strategy over the historical look-back window $[-\ell, 0]$. A higher Calmar ratio indicates that the strategy generates more expected return per unit of maximum drawdown.

Chapter 4

Climate Positioning

4.1 Defining climate positioning

In this section we introduce the concept of climate positioning. In mathematical finance, a position is defined with respect to a traded asset or vector of traded assets. Following Harrison and Pliska [27], let $S_t = (S_t^1, \dots, S_t^n)$ denote the vector of asset prices and $\theta_t = (\theta_t^1, \dots, \theta_t^n)$ the corresponding portfolio holdings. The portfolio value is then given by

$$V_t = \theta_t^\top S_t.$$

A position therefore represents the allocation of wealth across traded assets according to an investor's beliefs. Climate, however, is neither a traded asset nor a well-defined state variable. Instead, it must be represented by an appropriate climate state variable, denoted C_t , which may be constructed from tradeable assets, non-tradeable variables, or a combination of both. A climate position can therefore be viewed as a portfolio or derivative whose value depends on changes in C_t .

4.1.1 Harrison and Kreps (1978)

Harrison and Kreps [26] provide one of the foundational models of speculative trading under uncertainty by showing that rational investors with heterogeneous beliefs can generate speculative demand and influence equilibrium asset prices. Unlike earlier models, which assumed homogeneous expectations, they demonstrate that differences in subjective beliefs alone are sufficient to motivate trading.

In their framework, each investor evaluates future payoffs using their own subjective probability measure. If investor i assigns probability measure P_i to future states of the world, the expected payoff of a claim is $E_{P_i}[X]$, and in general $E_{P_i}[X] \neq E_{P_j}[X]$. Although investors observe the same information, they may assign different probabilities to future outcomes, leading them to value identical assets differently and creating speculative trading.

This framework naturally extends to climate finance, where investors often disagree about future climate outcomes, such as decarbonisation, climate policy and technological progress.

If C_T denotes a climate state variable, then differing beliefs imply

$$P_A(C_T) \neq P_B(C_T) \implies E_{P_A}[f(C_T)] \neq E_{P_B}[f(C_T)],$$

so identical climate-contingent claims may receive different valuations.

This dissertation extends the Harrison and Kreps framework by replacing uncertain future dividends with a climate state variable. Investors are assumed to speculate on future climate outcomes through derivative contracts with payoff

$$H = f(C_T),$$

thereby applying heterogeneous beliefs to climate-related uncertainty.

While Harrison and Kreps provide the economic motivation for speculative trading, they do not consider derivative pricing, arbitrage-free valuation or climate-related state variables. These omissions motivate the present work, which combines heterogeneous beliefs with modern derivative pricing to develop instruments that allow investors to express differing views on future climate outcomes. The next section therefore introduces the climate state variable on which these contracts are based.

4.1.2 Climate State Variable

The preceding chapter established the foundations of continuous-time asset pricing, where derivative payoffs depend on the future value of one or more traded assets. In the Black–Scholes framework, for example, the underlying state variable is the stock price process S_t . This dissertation instead first considered derivatives whose value depends on future climate outcomes, requiring the introduction of a suitable climate state variable. Unlike equities, commodities or interest rates, “climate” is not itself a traded asset but an evolving economic state. It may be represented by factors such as climate policy, carbon pricing, technological innovation and investor expectations.

The climate state variable is defined as an $\mathcal{F}_t$ -adapted stochastic process

$$C = \{C_t\}_{t \geq 0},$$

defined on the filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P),$$

where C_t represents the prevailing state of the climate transition at time t . Since $C_t \in \mathcal{F}_t$, the climate state depends only on information available up to time t . Changes in C_t therefore reflect the arrival of new information, such as climate policy announcements, carbon market developments and technological innovation. The precise interpretation of C_t will depend on the derivative being modelled. To embed the climate state within a continuous-time pricing framework, we assume that it evolves according to the stochastic differential equation

$$dC_t = \mu_C(C_t, t) dt + \sigma_C(C_t, t) dW_t,$$

where μ_C and σ_C denote the drift and volatility of the process, respectively, and W_t is a standard Brownian motion. The specific forms of μ_C and σ_C depend on the chosen specification of the climate state.

The climate state variable therefore plays the same mathematical role as the underlying asset in classical derivative pricing. Whereas a European option derives its value from the future asset price S_T , the proposed climate-positioning instruments derive their value from the future climate state C_T .

4.2 Climate Positioning (Tradeable underlying)

Having introduced the climate state variable C_t , climate positioning can now be defined depending on what we decide C_t to be within the standard portfolio-theoretic framework. If we do use a tradable asset in order to proxy the climate state variable then we can define the climate position as something more similar to our conventional financial position.

Let $C = \{C_t\}_{t \geq 0}$ denote an $\mathcal{F}_t$ -adapted climate state variable defined on the filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P).$$

Let V_t denote the time- t value of a portfolio or derivative contract whose value may depend on both conventional financial assets and the climate state variable. We write this dependence as

$$V_t = V(t, S_t, C_t),$$

where S_t denotes the vector of traded financial asset prices.

If the climate proxy is genuinely tradeable, the extension of the standard Harrison–Pliska framework is straightforward. Let the portfolio process be defined as

$$\theta_t = (\theta_t^0, \theta_t^1, \dots, \theta_t^n, \theta_t^C), \quad (4.1)$$

where θ_t^C denotes the holding in the climate instrument. The portfolio value is then given by

$$V_t = \theta_t^0 S_t^0 + \sum_{i=1}^n \theta_t^i S_t^i + \theta_t^C C_t, \quad (4.2)$$

where C_t denotes the price of the climate instrument. The climate position is therefore defined as

$$\Pi_t^C := \theta_t^C, \quad (4.3)$$

which represents the number of units of the climate instrument held. A climate-positioning derivative is therefore a contingent claim whose payoff depends explicitly on the future

realisation of the climate state variable. Formally, a derivative with maturity T is a climate-positioning instrument if its terminal payoff can be written as

$$H = f(C_T),$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a measurable payoff function. More generally, the payoff may depend jointly on financial and climate variables,

$$H = f(S_T, C_T).$$

The purely climate-contingent case $H = f(C_T)$ is the central focus of this dissertation. The economic interpretation of this definition is straightforward. An investor with a positive climate position benefits from an increase in the climate state variable, while an investor with a negative climate position benefits from a decrease. The meaning of “increase” depends on the construction of C_t . If C_t represents the strength of climate transition, then a long climate position corresponds to a position that gains when the market-implied transition accelerates. Conversely, a short climate position gains when transition expectations weaken.

4.2.1 Properties of the Climate State Variable

The climate state variable introduced in the previous section forms the mathematical foundation of the proposed climate-positioning framework. Consequently, it is necessary to specify the class of stochastic processes that are admissible within the present model. Throughout this dissertation, the climate state variable is assumed to satisfy the standard regularity conditions required for continuous-time asset pricing.

A stochastic process $C = \{C_t\}_{t \geq 0}$ is said to be an admissible climate state variable if:

- (a) C is $\mathcal{F}_t$ -adapted and measurable;
- (b) C has finite second moments, $\mathbb{E}[C_t^2] < \infty$ for all $t \geq 0$;
- (c) C takes values almost surely in a domain $\mathcal{D} \subseteq \mathbb{R}$ fixed in advance by the choice of proxy for C_t ($\mathcal{D} = \mathbb{R}$, $(0, \infty)$, or $[0, 1]$ as appropriate), and its dynamics are specified by an admissible diffusion (or transformed diffusion) on $\mathcal{D}$;
- (d) C is sufficiently regular that contingent claims of the form $H = f(C_T)$ are well defined and possess finite expectation under the chosen pricing measure.

Conditions (a) and (b) ensure that the climate state variable satisfies the mathematical assumptions required by stochastic calculus and continuous-time asset pricing. Condition (c), ensures that the process is consistent with the specific observable used to proxy C_t , and rules out, for example, modelling a prediction-market-implied probability with an unbounded diffusion. Condition (d) provides the economic interpretation of the process: since climate transition is not itself, in general, a traded asset.

4.3 Climate Positioning (non-Tradeable underlying)

When the climate state variable C_t cannot be directly traded as in the case of a physical climate index such as sea surface temperature considered by Little et al. [35], or a latent transition-risk factor observed only through proxies the replication argument underpinning the Harrison–Pliska framework [27] breaks down. Since no portfolio of traded assets can perfectly replicate an arbitrary climate-contingent payoff

$$H = f(C_T),$$

the market is said to be climatically incomplete with respect to C_t . The mathematical consequences of this incompleteness are fundamental: the equivalent martingale measure is no longer unique, derivative prices are no longer determined by no-arbitrage alone, and the climate position can no longer be identified with an actionable portfolio holding in the climate instrument itself. The theory of incomplete markets is developed rigorously in Harrison and Pliska [27] and Karatzas and Shreve [34], while its implications for derivative pricing are treated extensively in Björk [4] and Duffie [13].

Following Harrison and Pliska [27] and Karatzas and Shreve [34], the financial market

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P, S)$$

is said to be climatically incomplete with respect to the climate state variable C if there exists a square-integrable, $\mathcal{F}_T$ -measurable payoff

$$H = f(C_T)$$

such that no self-financing trading strategy

$$\theta_t = (\theta_t^0, \theta_t^1, \dots, \theta_t^n)$$

in the traded assets S satisfies

$$V_T(\theta) = H, \quad \text{almost surely under } P. \quad (4.4)$$

Harrison and Pliska [27] establish that market incompleteness of this kind is equivalent to the non-uniqueness of the equivalent martingale measure. Formally, define the set of equivalent martingale measures consistent with no-arbitrage on the traded assets S by

$$\mathcal{Q} := \left\{ Q \sim P : \frac{S_t^i}{S_t^0} \text{ is a } Q\text{-martingale for all } i = 1, \dots, n \right\}. \quad (4.5)$$

In a complete market,

$$|\mathcal{Q}| = 1,$$

and derivative prices are uniquely determined by the principle of no-arbitrage. In the climatically incomplete case, however, $\mathcal{Q}$ contains infinitely many equivalent martingale mea-

tures, each corresponding to a different assumption regarding the compensation investors require for bearing unhedgeable climate risk. As shown by El Karoui and Quenez [14], this non-uniqueness implies that the no-arbitrage price of a climate-contingent claim is not a single value, but instead lies within the interval bounded by the super-replication and sub-replication prices,

$$\underline{V}(t, C_t) \leq V(t, C_t) \leq \bar{V}(t, C_t). \quad (4.6)$$

Here, $\bar{V}(t, C_t)$ denotes the super-replication price, defined as the minimum cost of a self-financing portfolio in the traded assets that dominates the payoff H almost surely, while $\underline{V}(t, C_t)$ denotes the corresponding sub-replication price [14]. Any price lying within this interval is consistent with the absence of arbitrage in the traded assets S . Consequently, additional economic or modelling assumptions are required to select a unique pricing measure and hence a unique derivative price.

4.3.1 Climate Position in the Non-Tradeable Case

In the tradeable case, the climate position was defined as the holding θ_t^C in the climate instrument, a quantity that is directly actionable because the investor can physically hold units of the traded asset. In the non-tradeable case this definition is no longer available, since there is no climate instrument to hold directly. Instead, the climate position is defined in terms of the super-replicating portfolio, which is the closest analogue to a replicating portfolio available in an incomplete market [14]. Let

$$\bar{\theta}_t = (\bar{\theta}_t^0, \bar{\theta}_t^1, \dots, \bar{\theta}_t^n) \quad (4.7)$$

denote the super-replicating trading strategy in the traded assets S , that is, the minimum-cost self-financing portfolio satisfying

$$V_T(\bar{\theta}) \geq f(C_T), \quad \text{almost surely under every } Q \in \mathcal{Q}. \quad (4.8)$$

The value of the super-replicating portfolio at time t is

$$\bar{V}_t = \bar{\theta}_t^0 S_t^0 + \sum_{i=1}^n \bar{\theta}_t^i S_t^i, \quad (4.9)$$

and satisfies the self-financing condition [27]

$$d\bar{V}_t = \bar{\theta}_t^0 dS_t^0 + \sum_{i=1}^n \bar{\theta}_t^i dS_t^i. \quad (4.10)$$

Since no traded climate instrument exists, the portfolio process contains no climate holding analogous to θ_t^C in the complete-market framework. The best an investor can achieve using traded assets alone is the super-replicating portfolio $\bar{\theta}_t$, whose value is $\bar{V}_t$. The climate position in the non-tradeable case is therefore defined as

$$\Pi_t^C := \bar{V}_t - V(t, C_t) \geq 0. \quad (4.12)$$

The quantity Π_t^C is observable at time t , and it tells us how far short the traded price falls of what it would actually cost to hedge the payoff with certainty. Since (4.7) already tells us $\bar{V}_t \geq V(t, C_t)$, it follows immediately that $\Pi_t^C \geq 0$, and it is zero exactly when $H = f(C_T)$ can be replicated using traded assets alone, at which point we are back in the complete-market case. When Π_t^C is strictly positive, a seller who writes the derivative and hedges with the super-replicating portfolio $\bar{\theta}_t$ is left carrying an exposure of size Π_t^C : building the safe hedge costs more than what they were paid for the derivative in the first place. The seller takes this on because the price $V(t, C_t)$ reflects what the market is collectively willing to pay, and that need not equal the cost of eliminating the risk entirely. This is really the same idea as Harrison and Kreps [26]: a buyer with subjective beliefs $P^A(C_T)$ may value the payoff highly enough to transact at a price the seller cannot fully cover. In that sense the seller's residual position is long climate risk, a favourable move in C_T works in their favour, an unfavourable one against them, even though there is no instrument they could have held instead. Π_t^C is therefore the non-tradeable analogue of the holding θ_t^C from the previous section, standing in for it in a market where no such holding exists. Furthermore, while Π_t^C is defined without reference to an expectation, it can be evaluated using the pricing formula since

$$V(t, C_t) = e^{-r(T-t)} E^Q [f(C_T) | \mathcal{F}_t], \quad (4.11)$$

and

$$\bar{V}_t = e^{-r(T-t)} E^Q [V_T(\bar{\theta}) | \mathcal{F}_t], \quad (4.12)$$

it follows that

$$\Pi_t^C = e^{-r(T-t)} \mathbb{E}^Q [V_T(\bar{\theta}) - f(C_T) | \mathcal{F}_t]. \quad (4.15)$$

The expectation therefore appears only as a consequence of applying the pricing formula to evaluate Π_t^C ; it is not part of the definition itself. This is analogous to the tradeable case, where the sensitivity $\partial V / \partial C_t$ arises from applying Itô's lemma to the pricing function rather than constituting the primary definition of the climate position.

4.3.2 Climatic Incompleteness: The Market Price of Climate Risk

To assign a unique price to the derivative from within the no-arbitrage interval established by El Karoui and Quenez [14], it is necessary to select a particular equivalent martingale measure from $\mathcal{Q}$. Following Björk [4] and Duffie [13], this selection is parametrised by the market price of climate risk, λ_C , which governs the Girsanov transformation between the physical measure P and the chosen pricing measure Q .

Under the physical measure P , the climate state variable evolves according to

$$dC_t = \mu_C(C_t, t) dt + \sigma_C(C_t, t) dW_t^C, \quad (4.13)$$

where W_t^C is a standard Brownian motion under P .

Under the equivalent martingale measure Q , the dynamics become

$$dC_t = [\mu_C(C_t, t) - \lambda_C \sigma_C(C_t, t)] dt + \sigma_C(C_t, t) d\widetilde{W}_t^C, \quad (4.14)$$

where $\widetilde{W}_t^C$ is a standard Brownian motion under Q and $\lambda_C \in \mathbb{R}$ denotes the market price of climate risk.

The corresponding Girsanov transformation is

$$\left. \frac{dQ}{dP} \right|_{\mathcal{F}_T} = \exp \left(- \int_0^T \lambda_C dW_t^C - \frac{1}{2} \int_0^T \lambda_C^2 dt \right). \quad (4.15)$$

For this change of measure to define a valid probability measure, the Novikov condition must hold [48]; see also Karatzas and Shreve [34]:

$$E^P \left[\exp \left(\frac{1}{2} \int_0^T \lambda_C^2 dt \right) \right] < \infty. \quad (4.16)$$

When λ_C is constant or uniformly bounded, this condition is automatically satisfied.

In the tradeable case, no-arbitrage determines λ_C endogenously as the Sharpe ratio of the traded climate instrument [4],

$$\lambda_C = \frac{\mu_C - r_{C_t}}{\sigma_C}. \quad (4.17)$$

In the non-tradeable case, however, λ_C is a free parameter that must be specified exogenously by the modeller, since no traded climate instrument exists to determine it through no-arbitrage. Different choices of λ_C correspond to different equivalent martingale measures in $\mathcal{Q}$ and therefore produce different derivative prices, all of which remain consistent with the absence of arbitrage in the traded assets [14, 27].

Given a choice of λ_C , the derivative price is

$$V(t, C_t) = e^{-r(T-t)} E^Q [f(C_T) | \mathcal{F}_t], \quad (4.18)$$

and satisfies the pricing partial differential equation

$$\frac{\partial V}{\partial t} + (\mu_C - \lambda_C \sigma_C) \frac{\partial V}{\partial C} + \frac{1}{2} \sigma_C^2 \frac{\partial^2 V}{\partial C^2} - rV = 0, \quad (4.19)$$

subject to the terminal condition

$$V(T, C_T) = f(C_T). \quad (4.20)$$

The term

$$-\lambda_C \sigma_C \frac{\partial V}{\partial C}$$

represents the climate risk adjustment to the drift. It shifts the drift of the climate state variable from its physical value μ_C to the risk-neutral value $\mu_C - \lambda_C \sigma_C$ and vanishes when $\lambda_C = 0$. This adjustment is structurally analogous to the volatility risk premium appearing

in stochastic volatility models [4]. Consequently, the pricing equation has the same mathematical structure as the Black–Scholes equation, with the important distinction that σ_C is the volatility of a non-traded state variable and λ_C cannot be inferred from market prices but must instead be specified as part of the pricing model.

4.4 Climate positioning for a QIS

For the purpose of creating a climate position for the case of a QIS, climate information must first be represented by a measurable quantity and then translated into exposure to traded financial assets. This motivates defining climate positioning for this case through the mapping from observed climate information into portfolio holdings.

A useful precedent is provided by Engle et al. [15]. Rather than treating climate as a tradeable underlying, they construct a measurable climate-news series, denoted CC_t , and form portfolios of traded securities whose returns are exposed to innovations in this series. Their construction therefore provides an important intuition for the present framework: a non-tradeable source of climate risk can be expressed financially through positions in tradeable assets. The QIS developed in this dissertation applies the same principle to physical climate variables rather than textual climate news.

Let $C = (C_t)_{0 \leq t \leq T}$ denote a climate state as usual in this chapter, such as temperature or rainfall, with natural filtration $\mathcal{F}_t$.

4.4.1 Climate signal.

A climate signal is an $\mathcal{F}_t$ -adapted process

$$\xi_t = \xi(t, (C_s)_{s \leq t}), \quad (4.21)$$

taking values in a signal space $\mathcal{X}$, for example

$$\mathcal{X} = \{-1, 0, 1\} \quad \text{or} \quad \mathcal{X} = \mathbb{R}.$$

The signal is a measurable summary of the climate path observed up to time t .

4.4.2 Climate positioning rule.

Let

$$S_t = (B_t, S_t^1, \dots, S_t^n)$$

denote the vector of traded assets. A climate positioning rule is a measurable map

$$g : [0, T] \times \mathcal{X} \rightarrow \mathbb{R}^{n+1} \quad (4.22)$$

such that

$$\theta_t = g(t, \xi_{t-}) \quad (4.23)$$

defines a predictable, admissible and self-financing trading strategy, with

$$V_t = \theta_t^\top S_t, \quad dV_t = \theta_t^\top dS_t. \quad (4.24)$$

Climate enters the portfolio only through the signal ξ ; the strategy does not require the climate variable itself to be tradeable. Using ξ_{t-} ensures that positions are based only on information available before the trade is formed, thereby ruling out look-ahead.

4.4.3 Climate position.

Let $\xi^0 \in \mathcal{X}$ denote the neutral signal and define the corresponding neutral allocation as

$$\theta_t^0 = g(t, \xi^0). \quad (4.25)$$

The climate position is then

$$\Pi_t^C := \theta_t - \theta_t^0 = g(t, \xi_{t-}) - g(t, \xi^0), \quad (4.26)$$

representing the component of the portfolio held specifically because of climate information. If the neutral allocation is flat, as in Chapter 7, then $\Pi_t^C = \theta_t$. Climate positioning can therefore be represented by the triple

$$\boxed{(C, \xi, g)}, \quad (4.27)$$

where C is the climate state, ξ extracts an investable signal from that state, and g maps the signal into holdings in liquid traded instruments.

In the coffee QIS later in this dissertation, for example, temperature conditions generate the signal, the signal determines whether a position is taken, and the resulting position is implemented through Arabica coffee futures. The strategy is therefore a climate position because its financial exposure is systematically conditional on observed climate information.

Chapter 5

Data analysis

In this chapter, our aim is to investigate the price of coffee in order to see what has affected it in the past. More specifically, we want to see whether price changes have occurred as a direct implication of climate events. We choose to investigate this commodity because coffee is a widely consumed commodity which is also very sensitive to physical changes in climate. All the data analysis will be done using Python or R.

5.1 Coffee introduction and supply chain

Coffee primarily comes in two types, that being Arabica and Robusta. Arabica accounts for approximately 60–70% of global coffee production and is generally regarded as a higher-quality bean with a smoother, more complex flavour profile and lower caffeine content. It is typically cultivated at higher altitudes (600–2,000 m) in cooler climates and is particularly sensitive to changes in temperature, rainfall, pests, and disease. In contrast, Robusta is more resilient to environmental stress, exhibiting greater tolerance to higher temperatures, lower elevations, and variable rainfall conditions. Robusta beans contain approximately twice the caffeine content of Arabica, resulting in a stronger, more bitter flavour, and are widely used in instant coffee and espresso blends due to their lower production costs and hardier growing characteristics. These biological and agronomic differences imply that climate change is expected to affect the two species differently, with Arabica generally considered to be substantially more vulnerable to rising temperatures and changing precipitation patterns than Robusta.

An interesting paper by Felice(2025) [19] summarises and talks about the coffee supply chain very well and gives a great overview of the commodity and how everything related to it works. The coffee supply chain is one of the most complex and globalised supply chains out there involving millions of farmers, exporters, producers and consumers. The coffee supply chain consists of four main phases: Cultivation, processing, Roasting, and Consumption [8]. At every step within these four stages things like climate, labour and market demands effect the interactions between stakeholders.

Within the first stage of the coffee supply chain (cultivation) farmers are responsible for

selecting the coffee varieties, preparing land and planting seedlings, applying fertiliser, managing irrigation, and finally harvesting the grown crops. Table 5.1 shows us the main harvesting and flowering periods for some of the major coffee players. The data was sourced from individual coffee reports on the USDA website [54]. For example for Brazil [63] the report goes into detail about the trends and information related to coffee recently. On the main website there are many such reports like this one for the other big coffee producing countries. Within the 4 stages climate change and climate related news effect growers the most by far.

Table 5.1: Coffee harvest and growing seasons by country

Country	Species (harvests/year)	Main harvest	Secondary harvest	Flowering/Growing
Brazil [63]	Arabica & Robusta (1)	May–September	None	Flowering Sep–Nov
Vietnam [67]	Robusta (1)	October–January	None	Rainy season May–Sep
Colombia [57]	Arabica (2)	September–December	April–June	Flowering varies by region
Indonesia (low) [66]	Robusta (1)	March–May	None	Wet season growth
Indonesia (high) [66]	Robusta (1)	June–August	None	Wet season growth
Ethiopia [64]	Arabica (1)	October–January	None	Flowering Mar–Apr
Honduras [58]	Arabica (1)	November–March	None	Flowering Apr–May
Peru [61]	Arabica (1)	April–September	None	Flowering Sep–Nov
Uganda [56]	Robusta (2)	October–January	April–June	Rain-driven
India [65]	Arabica & Robusta (1)	November–February	None	Monsoon growing season
Mexico [60]	Arabica (1)	October–March	None	Flowering Apr–May

Table 5.2: Top ten coffee-producing countries (2025/26)

Rank	Country	Production (Million 60 kg Bags)
1	Brazil	63.00
2	Vietnam	30.80
3	Colombia	13.80
4	Indonesia	12.45
5	Ethiopia	11.56
6	Uganda	6.88
7	India	6.05
8	Honduras	5.80
9	Peru	4.20
10	Mexico	3.90

Table 5.2 shows the top countries when it comes to producing coffee [55]. We can see here that Brazil is easily the biggest coffee market in terms of total production with Vietnam being the biggest robusta producer in the world. This is also evident from the historical plots as shown in figure 5.1.

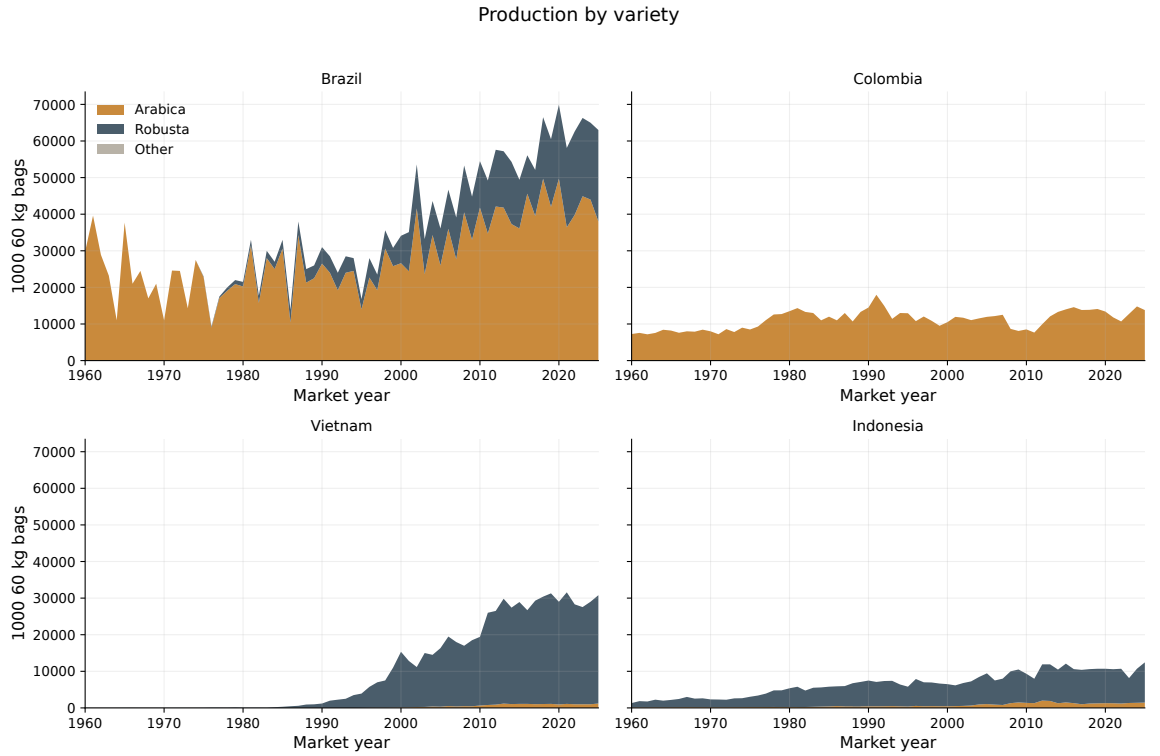


Figure 5.1: Annual coffee production by country and variety.

The next phase with reference to the earlier process we mentioned from the paper by Felice [19] is the processing stage. This is the stage in which the coffee is the processing

stage in which the beans are processed by either dry methods or wet methods [1] and [24]. The processed beans are transported to dry mills for sorting and testing, then to warehouses where the beans are often graded and then ready for exports.

The next stage within the supply chain is the roasting stage whereby here roasting transforms the green coffee beans into aromatic roasted brown beans used to produce the coffee beverages people drink. During roasting the beans are heated to approximately between 180 degrees celcius and 240 degrees celcius. [9]. Roasting is usually done in the consuming countries rather than the producing countries.

The final stage of the coffee supply chain is the consumption stage, during which coffee is prepared and consumed in cafés, restaurants, workplaces, and homes. Consumer preferences strongly influence the demand for different coffee varieties and products. The top coffee consumers are shown in table 5.3. The data was sourced from the united states department of agriculture [55]. We will examine trends a bit more in the next section.

Table 5.3: Top ten coffee-consuming markets (2025/26)

Rank	Country/Region	Consumption (Million 60 kg Bags)
1	European Union	41.87
2	United States	26.55
3	Brazil	22.28
4	Philippines	6.75
5	Japan	6.72
6	China	5.85
7	Canada	5.20
8	Vietnam	4.90
9	Indonesia	4.81
10	Russia	4.58

Its not surprising to see that the countries which produce the most coffee also export the most coffee. Table 5.4 shows this with the obvious players like Brazil, Vietnam, Columbia, and Indonesia being here. Although Brazil has the largest coffee market among producing countries, Vietnam, Honduras and Uganda export the majority of their production. For example Brazil produces approximately 63 million bags of coffee and consumes 22 million of that domestically whilst exporting or storing the rest meaning that Brazil has both a big domestic and international market. This is different to Vietnam who produces approximately 31 million bags of coffee but consumes only 5 million of that domestically meaning that a significant portion of their production is sold overseas. Consequently, a climate related shock to lets say Vietnam would hurt both Vietnam’s economy and the individuals who rely on their goods.

Table 5.4: Top ten coffee-exporting countries and regions (2025/26)

Rank	Country/Region	Exports (Million 60 kg Bags)
1	Brazil	40.75
2	Vietnam	27.90
3	Colombia	12.55
4	Indonesia	8.87
5	Ethiopia	7.80
6	Uganda	6.52
7	India	5.99
8	Honduras	5.50
9	European Union	5.50
10	Peru	4.38

Likewise, the top coffee consuming countries tend to be countries that import coffee the most seen as though these countries tend not to produce their own coffee. Table 5.5 shows this. something interesting to see here is that the European Union is 9th in exporting as shown in 5.4 but this is most likely simply due to the fact that the EU imports so much coffee that there is left over supply which might as well get sold if its not suitable to store every little bit.

Table 5.5: Top ten coffee-importing countries and regions (2025/26)

Rank	Country/Region	Imports (Million 60 kg Bags)
1	European Union	47.00
2	United States	26.50
3	Japan	6.55
4	Philippines	6.30
5	China	5.40
6	Canada	5.20
7	Russia	4.58
8	United Kingdom	4.33
9	Switzerland	3.60
10	South Korea	3.40

Another interesting table in the USDA page gives us information on the ending stocks for each major market participant. Ending stocks represent the quantity of coffee remaining in storage at the end of each marketing year and therefore acts as a buffer against future supply shocks. USDA data [55] and table 5.6 indicate that global ending stocks declined from 31.9 million 60 kg bags in 2020/21 to 20.1 million bags in 2025/26, suggesting that the global coffee market has become progressively tighter. As inventories diminish, the ability of producers and traders to offset climate-induced production shortfalls is reduced, increasing the likelihood of supply shortages and price volatility.

Table 5.6: Global coffee ending stocks by marketing year

Marketing Year	Ending Stocks (Million 60 kg Bags)
2020/21	31.94
2021/22	26.93
2022/23	23.12
2023/24	21.31
2024/25	22.82
2025/26	20.15

5.1.1 Consumption trends and links to supply chains

Carrying on from the same paper [19] the next section speaks about how recent shifts in coffee consumption patterns are significantly reshaping the structure and behaviour of the coffee supply chain. It is becoming increasingly evident that consumers take into account sustainability, transparency, origin, and ethical sourcing in their coffee purchasing decisions. Surveys show that over 60 percent of consumers in North America and Europe prefer coffee certified as organic and fairtrade, people are also willing to pay a premium for such things. This shift in demand has a direct impact on the production process as the companies involved in the growing process will need to adopt organic farming methods and attain fair trade certificates if they wish to continue to compete. Exported oriented countries like Ethiopia, Colombia, and Vietnam are trying hard to adopt these kinds of methods. Exporters and roasters are investing in direct trade partnerships in order to eliminate intermediaries and provide consumers with greater visibility into farm-level practices. The paper also mentions that the digitalisation of the supply chain is also a big response to this trend. Apps utilising QR based traceability allow consumers to scan and view the entire journey of their coffee, down to the farmer who harvested it. Different parts of the world express interest for different things, Europe emphasizes ethical sourcing and environmental impact, whereas the Asia - pacific markets show increased interest in quality, brewing innovations, and flavour. North American consumers like having subscriptions, transparency, and third-wave coffee culture which combined with the preferences of all the other continents reshapes the logistics and marketing behind coffee production.

5.2 Climate effect on coffee

It is obvious climate conditions directly affect coffee cultivation and can therefore influence market prices through changes in yield, quality and supply. This section examines the extent to which climate risk has affected coffee prices and whether it provides useful predictive signals.

Engle et al. (2020) [15] use the Wall Street Journal climate news index to construct a climate-change hedge portfolio. A similar approach can be applied here by using climate news as a measure of climate risk and testing its relationship with coffee prices alongside conventional economic factors. An advantage of the index is that it distinguishes between physical and transition risk, allowing the main channel of climate exposure to be identified.

Ma et al. (2024) [37] apply related methods to agricultural commodity returns. This chapter therefore draws on both studies, together with evidence from the wider literature, to assess whether climate risk helps explain movements in coffee prices.

The original WSJ climate news index is no longer publicly available. However, the climate-risk series used by Ma et al was obtained from Renato Faccini’s website [16]. This allows their empirical approach to be adapted to coffee price data sourced from LSEG Workspace.

Explaining Faccinis news index

The news index developed by Faccini will play a role in how the rest of this chapter goes. Faccini created this news index as part of the paper “Dissecting Climate Risks: Are they reflected in stock prices” (2023) [17]. This was also talked about in the literature review within this dissertation.

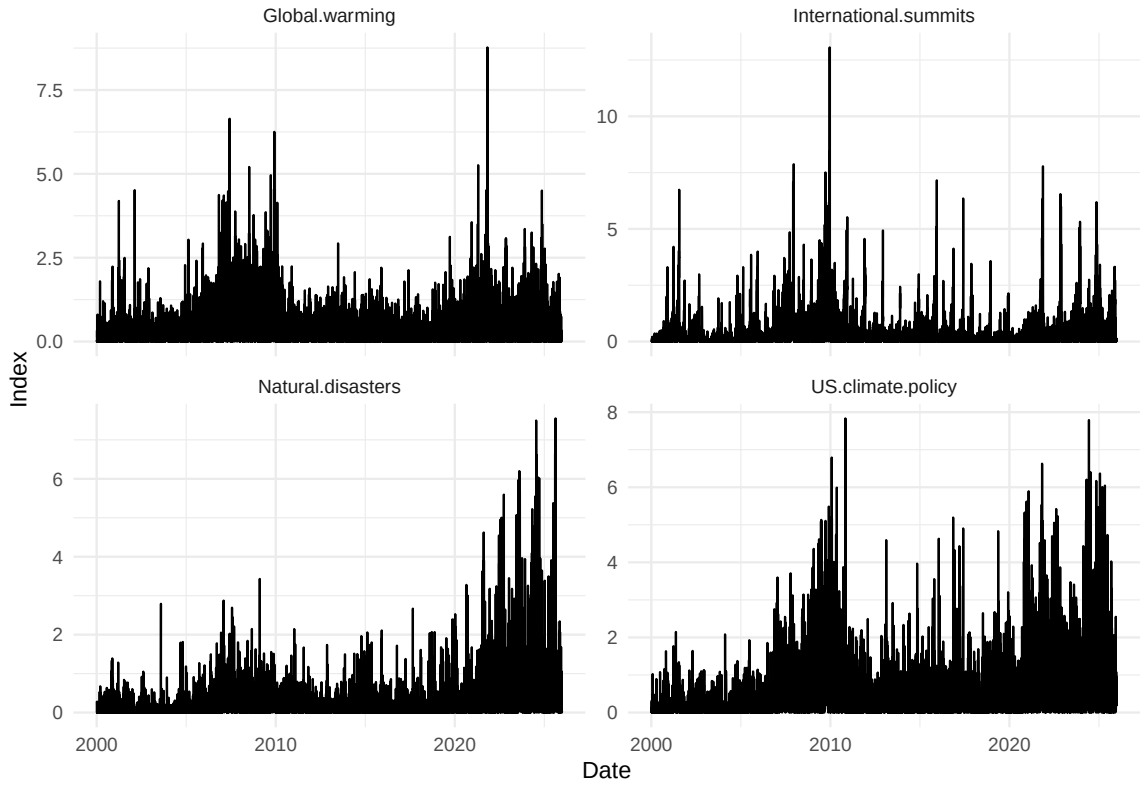


Figure 5.2: Plot of different news indexes

Figure 5.2 shows the plots for the different columns within the news index, the spikes show significant climate news events relative to the given category at the time. Table 5.7 shows the summary statistics for the original data before any transformations.

Table 5.7: Summary Statistics of Climate News Indices

	Mean	1st	Median	99th	Std. Dev.	Skew	$\rho(1)$
US Climate Policy	0.69	0.00	0.38	4.18	0.87	2.40	0.49
International Summits	0.41	0.00	0.11	4.00	0.80	4.69	0.65
Global Warming	0.51	0.00	0.30	2.92	0.64	2.84	0.49
Natural Disasters	0.37	0.00	0.11	3.50	0.69	3.83	0.57

From these numbers in 5.7 we can see that all the categories have a positive number in the skewness column, meaning that most of the data points are clustered to the left side of the distribution which indicates that extremely high levels of climate risk are relatively rare. All four climate news indices have relatively low average values but considerably higher 99 percentiles which suggests that climate related news is generally at low levels but occasionally experiences large spikes during major climate news events, policy announcements or international conferences. The median is also relative low, lower than mean which indicates that distributions are asymmetric, with a few periods of exceptionally high news intensity pulling the average upwards. This is evident visually anyway through the plots as there clearly are obvious large spikes in all the categories.

The US climate policy channel has the highest average at 0.69 which indicates that US climate policies have the greatest sustained media attention among the four topics, with occasional large spikes corresponding to major policy announcements. The international summits channel has a lower average with a mean of 0.41 but has the highest autocorrelation out of the four channels at a value of 0.65. This makes sense because international summits do not occur frequently but when they do the effects from the topics discussed persist for longer times. Global warming gets a decent amount of media coverage as the mean is 0.51 but it also has the lowest standard deviation at a value of 0.64. This makes sense as global warming is not something that causes extreme variation but rather has a slower but longer term effect. Finally, the natural disasters channel has the lowest average among the four channels seen as though extreme weather events happen infrequently especially in comparison to these other factors, but when these events do occur there is still a sizeable spike in media attention.

Following on from the methodology of Ma's paper [37] we decide to perform PCA on the data in order to get an aggregate for climate news as it helps to filter out some of the noise and makes it easier when it comes to later sections involving decomposing the price of coffee into the variables it is explained by. We can also test to see for sure if the individual components have better predictive power or if the aggregated information has better predictive power before choosing what to test alongside the other economic predictors.

Analysis of coffee with climate news index

The first objective is to work out the forecasting power of individual predictors. We will achieve this by fitting a simple uni-variate regression using the same methodology as in

previous literature (e.g., Goyal and Welch, 2008; Gargano and Timmermann, 2014) [20, 23]. To be very specific we are using monthly US arabica coffee futures data (KCC1) from LSEG.

The univariate return regression as in [37] is represented as:

$$R_{t+1:t+h} = \alpha + \beta CR_t + \varepsilon_{t+1:t+h} \quad (5.1)$$

where $R_{t+1:t+h}$ is the cumulative return from $t + 1$ to $t + h$, CR_t is the predictor variable at time t , α is the intercept, β is the slope coefficient, and $\varepsilon_{t+1:t+h}$ is the error term. We will have three forecast horizon times, that being 1 month, 6 months and 12 months. CR_t refers to the climate change-related predictors (which in our case is the US climate policy, International summits, Global warming, Natural Disasters, and the PCA aggregated news). In order to perform the analysis we first needed to do some background data management to prepare everything. First of all we needed to create the initial merged data table containing the close, log returns, policy, summits, global warming, natural disasters, and PC1). The log returns were computed using this formula

$$R_t = 100 \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (5.2)$$

We chose to use monthly data for each column as its less noisy than using daily data as well as the fact that in Faccinis news index there were missing data points so in order to make sure everything aligned correctly it was better to average everything over a month. Next we compute the forecasting horizons as mentioned earlier in the paper, we do the 1 month horizon, 6 month horizon, and the 12 month horizon. Then we compute regressions for each of the different horizons getting the beta, P value, R squared and the NW statistic.

Table 5.8: Predictive regression results across forecasting horizons.

Predictor	1-month			6-month			12-month		
	β	NW- t	R^2 (%)	β	NW- t	R^2 (%)	β	NW- t	R^2 (%)
PC1	1.0112	2.29	1.22	4.2656	2.35	4.12	7.9026	2.18	6.34
US Climate Policy	1.0653	2.47	1.35	4.8809	2.36	5.42	8.7455	1.94	7.32
International Summits	0.3993	1.00	0.19	1.5036	1.54	0.51	4.1925	1.67	1.80
Global Warming	1.1728	2.78	1.63	3.2119	2.19	2.35	6.9185	2.19	4.95
Natural Disasters	0.3875	0.66	0.18	4.2116	1.65	3.67	5.0271	1.29	2.32

5.2.1 In-sample Predictive Regression Results

Table 5.8 reports the in-sample predictive regression results for the 1, 6, and 12 month forecasting horizons. Following Ma et al. (2024) [37], all climate-related predictors were standardised to have a mean of zero and a unit standard deviation prior to estimation. Consequently, the estimated coefficients can be interpreted as the change in future coffee futures returns associated with a one-standard-deviation increase in each climate risk

measure.

The principal component climate risk index (PC1) exhibits statistically significant predictive ability across all forecasting horizons. At the one-month horizon, the estimated coefficient is 1.0112 with a Newey–West t -statistic of 2.29, indicating statistical significance at conventional levels. Economically, this suggests that a one-standard-deviation increase in the composite climate risk index predicts approximately a 1.01 percentage point increase in the following month’s coffee futures return.

The predictive effect of the composite climate risk index strengthens over longer forecasting horizons. The estimated coefficient increases to 4.2656 at the six-month horizon and 7.9026 at the twelve-month horizon, while remaining statistically significant with Newey–West t -statistics of 2.35 and 2.18, respectively. Similarly, the corresponding in-sample R^2 values increase from 1.22% at the one-month horizon to 4.12% at the six-month horizon and 6.34% at the twelve-month horizon, indicating that climate risk explains a progressively larger proportion of the variation in longer-horizon coffee futures returns.

5.2.2 Individual Climate Risk Measures

Among the individual climate risk proxies, US Climate Policy demonstrates the strongest and most consistent predictive ability. The estimated coefficient increases from 1.0653 at the one-month horizon to 4.8809 at the six-month horizon and 8.7455 at the twelve-month horizon. The associated Newey–West t -statistics remain close to or above 2 across all forecasting horizons, suggesting statistically significant predictive power. Furthermore, US Climate Policy achieves the highest in-sample R^2 values among the individual predictors, reaching 7.32% at the twelve-month horizon.

Global Warming also exhibits strong predictive ability. The estimated coefficient rises from 1.1728 at the one-month horizon to 6.9185 at the twelve-month horizon, with Newey–West t -statistics exceeding 2 across all horizons. The explanatory power similarly increases over time, with the R^2 statistic rising from 1.63% at the one-month horizon to 4.95% at the twelve-month horizon. These findings suggest that global warming-related news contains economically meaningful information regarding future coffee futures returns.

By contrast, International Summits displays relatively weak predictive performance. Although the estimated coefficients remain positive across all forecasting horizons, the Newey–West t -statistics at the one-month and six-month horizons (1 and 1.54), respectively) do not indicate statistical significance at conventional levels. Only at the twelve-month horizon does the predictor become marginally significant ($t = 1.67$), while the corresponding R^2 remains comparatively low at 1.80%.

Similarly, Natural Disasters exhibits limited predictive ability. Although the estimated coefficients increase with the forecasting horizon, the associated Newey–West t -statistics (0.66, 1.65, and 1.29) remain a little low.

5.2.3 Overall Discussion

Overall, the results indicate that climate-related news possesses meaningful predictive power for coffee futures returns, particularly over medium and long-term forecasting horizons. Consistent with the broader climate finance literature, the explanatory power of the predictive regressions generally increases with the forecasting horizon, suggesting that climate-related information is incorporated gradually into coffee futures prices rather than immediately.

Among the individual climate risk measures, US Climate Policy emerges as the strongest predictor of future coffee futures returns, followed closely by Global Warming. In contrast, International Summits and Natural Disasters exhibit substantially weaker predictive ability throughout the sample period. However it should be noted that we are looking over a large time frame so natural disasters effects may get smoothed, but if we were to zoom in maybe we could observe predictive capability. The statistically significant performance of the composite principal component index further suggests that aggregating multiple dimensions of climate risk into a single indicator provides a useful summary measure for forecasting future coffee futures returns.

5.2.4 Testing against other economic predictors

We now want to see if the climate indices we created actually do provide climate related information or if it just mimics the information provided by other financial predictors. We will again be using a similar approach to Ma et als paper [37].

$$R_{t+1:t+h} = \alpha + \beta \text{PC-CR}_t + \psi E_t + \varepsilon_{t+1:t+h}. \quad (5.3)$$

where E_t is the economic predictors which in our case will not be exactly the same as Ma's [37] paper.

For the DXY data we used LSEG and for CPIAUCSL we used the Federal Reserve bank of ST.LOUIS (FRED). As for the 2025-10-01 entry used linear interpolation (taking average of adjacent months) as this month the FED did not release a number. For all the other variables we used FRED to get the data.

Table 5.9: Predictive regressions with economic controls: 1-month horizon

Control	$\beta_{\text{PC-CR}}$	NW-t	p-value	β_{Control}	NW-t	R^2 (%)	ΔR^2
TBL	1.0500	2.35	0.0195	-0.3792	-0.87	1.41	1.35
KI	0.7401	1.51	0.1315	0.8047	1.63	1.99	0.62
IP	0.9759	2.18	0.0301	1.5278	2.99	4.23	1.19
INFL	0.9670	2.17	0.0306	0.2938	0.69	1.35	1.16
GS10	1.0006	2.25	0.0253	-0.1427	-0.31	1.27	1.25
DXY	0.9755	2.11	0.0358	-0.2080	-0.41	1.30	1.17

Table 5.10: Predictive regressions with economic controls: 6-month horizon

Control	$\beta_{\text{PC-CR}}$	NW- t	p -value	β_{Control}	NW- t	R^2 (%)	ΔR^2
TBL	4.5097	2.41	0.0164	-1.6726	-0.66	4.74	4.51
KI	3.5959	1.79	0.0750	2.2041	0.93	5.13	2.66
IP	4.1580	2.27	0.0238	1.6299	1.92	4.58	3.93
INFL	4.2293	2.36	0.0190	-0.4537	-0.31	4.02	4.02
GS10	4.2779	2.32	0.0209	-0.5345	-0.21	4.19	4.15
DXY	4.2016	2.20	0.0286	-0.6803	-0.26	4.23	3.97

Table 5.11: Predictive regressions with economic controls: 12-month horizon

Control	$\beta_{\text{PC-CR}}$	NW- t	p -value	β_{Control}	NW- t	R^2 (%)	ΔR^2
TBL	8.1168	2.19	0.0295	-1.6434	-0.30	6.60	6.57
KI	7.1725	1.85	0.0653	2.2809	0.55	6.82	4.70
IP	7.6926	2.13	0.0342	1.4636	0.99	6.32	6.06
INFL	8.1187	2.42	0.0163	-3.8592	-1.63	7.64	6.68
GS10	7.9074	2.15	0.0322	-0.4840	-0.09	6.36	6.35
DXY	8.0737	2.14	0.0328	1.5392	0.30	6.58	6.54

The previous analysis established that the principal component climate risk index (PC-CR) possesses significant predictive power for coffee futures returns across multiple forecasting horizons. However, climate-related risks may also be reflected in broader macroeconomic conditions, including monetary policy, inflation, industrial production, exchange rates and global economic activity. Consequently, it is necessary to determine whether the predictive ability of PC-CR merely captures information already contained in conventional macroeconomic variables.

To examine this issue, six widely used macroeconomic variables are incorporated individually into the predictive regressions: the U.S. Dollar Index (DXY), inflation (INFL), the three-month Treasury bill rate (TBL), the ten-year Treasury yield (GS10), industrial production (IP), and the Kilian (2009) global real economic activity index (KI). Following Ma et al. (2024), each economic variable is entered separately alongside PC-CR in order to evaluate whether climate risk provides incremental predictive information beyond established macroeconomic fundamentals. The data for DXY came from LSEG workspace and from FRED [18] for the others.

Tables 5.9–5.11 present the estimation results for the one-, six-, and twelve-month forecasting horizons. Overall, the results indicate that the predictive ability of PC-CR remains largely unaffected after controlling for conventional macroeconomic variables. Across almost all specifications, the estimated coefficient on PC-CR remains positive and statistically significant, suggesting that climate-related information contains forecasting content that is not fully captured by traditional macroeconomic indicators.

At the one-month forecasting horizon, the estimated coefficients on PC-CR range from 0.74 to 1.05. The coefficient remains statistically significant at the 5% level when controlling

for the Treasury bill rate, industrial production, inflation, the ten-year Treasury yield and the U.S. Dollar Index, with Newey–West adjusted t -statistics ranging from 2.11 to 2.35. The only exception is the Kilian index, where the coefficient remains positive but is not statistically significant ($p = 0.1315$). These findings suggest that climate risk captures short-term movements in coffee futures returns beyond those explained by conventional macroeconomic conditions.

The explanatory power of the regressions also increases after incorporating PC-CR. Model R^2 values range between 1.27% and 4.23%, while the incremental explanatory power (ΔR^2) varies between 0.62% and 1.35%. The largest improvement is observed when TBL is included as the control variable, indicating that climate risk contributes additional predictive information beyond fluctuations in real economic activity.

The predictive performance of PC-CR strengthens considerably at the six-month forecasting horizon. Estimated coefficients increase to between 3.60 and 4.51, while Newey–West t -statistics exceed two in five of the six model specifications. Similar to the one-month results, the Kilian index represents the only specification in which statistical significance weakens slightly ($p = 0.0750$). Overall, these findings indicate that climate risk becomes increasingly informative for forecasting medium-term coffee futures returns.

The strongest evidence emerges at the twelve-month forecasting horizon. Estimated coefficients on PC-CR range between 7.17 and 8.12, with statistically significant Newey–West t -statistics in all specifications except the regression controlling for the Kilian index. Compared with the shorter forecasting horizons, the coefficient estimates increase steadily, indicating that climate risk becomes progressively more important when predicting longer-term coffee futures returns.

The explanatory power of the models is likewise highest at the annual horizon, with R^2 values ranging from 6.32% to 7.64%. Furthermore, the incremental explanatory power attributable to PC-CR exceeds six percentage points in most specifications, demonstrating that climate risk continues to provide substantial forecasting information even after controlling for exchange rates, inflation, interest rates, industrial production and global economic activity.

An additional observation concerns the macroeconomic control variables themselves. Although several controls exhibit economically intuitive coefficient signs, relatively few are statistically significant once climate risk is included in the predictive regressions. In contrast, the coefficient on PC-CR remains consistently significant across almost all model specifications. This finding suggests that the information contained within the climate risk index is not merely a reflection of conventional macroeconomic conditions but instead captures an independent source of variation in coffee futures returns.

Overall, these findings are consistent with the evidence presented by Ma et al. (2024). While macroeconomic variables undoubtedly influence commodity markets, they do not eliminate the predictive ability of climate risk. Instead, the results suggest that climate-related in-

formation provides incremental forecasting power beyond traditional macroeconomic fundamentals, particularly over medium- and long-term forecasting horizons. This motivates the use of climate related signals when it comes to crating a strategy for coffee futures.

5.3 Frost effect

Having shown that climate does indeed have an affect on coffee, we move on to see if we can extract some more event specific signals related to climate that affects the price of coffee. Fundamentally speaking all markets are driven by supply and demand. Supply relates to production levels and demand relates to how much people want the product. Now assuming that demand remains at a constant level a shock to the supply at this same level of demand would cause price to increase. This means that the level of production of a good does have an impact on the price of said good. If anything the level of supply has an impact on not only the price but also the forward curve of the asset due to the theory of storage, and the market conditions discussed in the literature review (contango and backwardation).

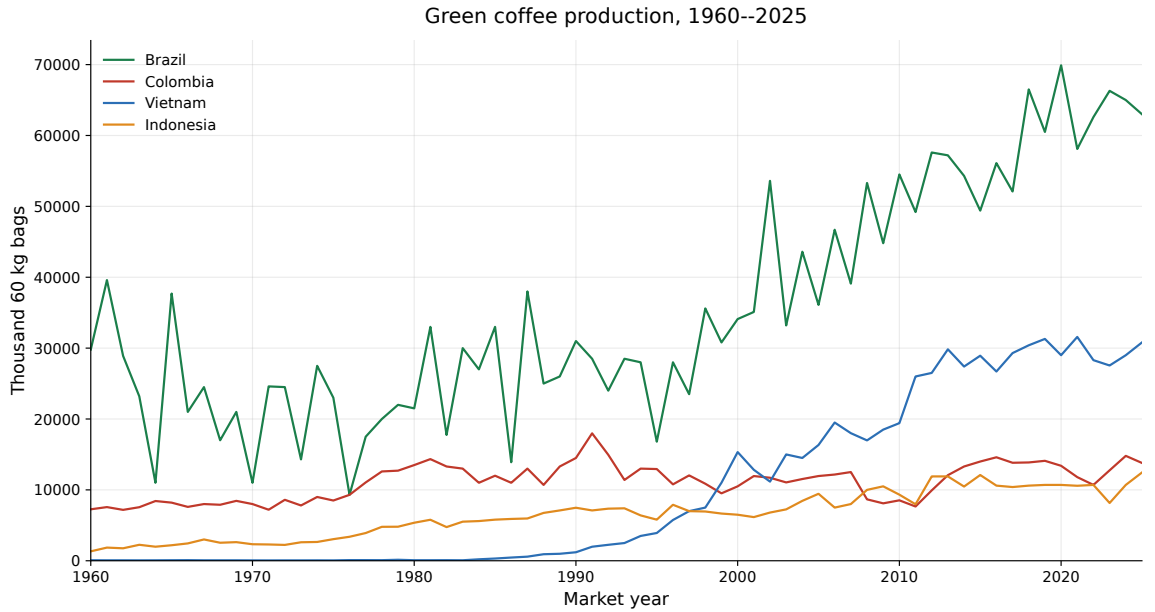


Figure 5.3: Historical production for major coffee-producing countries.

We can look at historical production levels from figure 5.3 and investigate why some periods had lower levels of production relative to its neighboring period. After a little bit of research it became clear that especially for Brazilian coffee Frost had an effect on coffee prices. There were 3 notable "black frosts" within the period of data we had for total production (1960-2026) and each "black frost" had devastating impacts on coffee production. This led to significant changes in price which motivates the reason to investigate frost further.

A particularly relevant study which was briefly mentioned in the literature review is that of Boudoukh et al. [7], who demonstrate that the relationship between orange juice futures

prices and frost is highly nonlinear. A similar mechanism operates in the coffee market, where Brazilian frost events have historically had substantial effects on both production and prices. For example, the severe frost of 1994 is estimated to have destroyed 30–40% of the following year’s crop. More recently, the July 2021 frost caused Arabica futures prices to increase by approximately 25% within a week and nearly 48% over the course of the year. While most of the 2021 harvest had already been completed when the frost occurred, the USDA notes that market participants anticipated damage to coffee trees and lower future production, leading to a sharp repricing of expected supply [53]. Subsequent harvests confirmed these expectations, with production remaining depressed for several years. Motivated by these episodes, we first examine whether historical frost events are systematically reflected in coffee futures prices by analysing returns across different minimum-temperature buckets.

after doing this we get a table that shows:

$$T_d^{\text{Brazil}} = \min(T_d^{\text{MG}}, T_d^{\text{PR}}, T_d^{\text{SP}}) \quad (5.4)$$

Table 5.12: Daily Arabica futures returns by minimum temperature across Minas Gerais, Paraná, and São Paulo during the Brazilian frost season (May–September)

Temp. Bucket	Obs.	Mean Return	Std. Dev	Obs. Share	Var. Share
$\leq 0^\circ\text{C}$	12	4.246	10.659	0.311	6.148
$0\text{--}2^\circ\text{C}$	26	-0.177	5.712	0.674	3.420
$2\text{--}5^\circ\text{C}$	126	-0.618	2.958	3.264	4.773
$5\text{--}10^\circ\text{C}$	1094	-0.083	2.487	28.342	28.334
$> 10^\circ\text{C}$	2602	0.024	2.294	67.409	57.326

Table 5.13: Comparison of observations across temperature buckets using alternative regional temperature measures

Temperature Bucket	Minas Gerais	Paraná	São Paulo	Combined Brazil
$\leq 0^\circ\text{C}$	0	12	1	12
$0\text{--}2^\circ\text{C}$	3	26	0	26
$2\text{--}5^\circ\text{C}$	23	122	17	126
$5\text{--}10^\circ\text{C}$	763	760	253	1094
$> 10^\circ\text{C}$	3071	2940	3589	2602

Table 5.12 shows that mean coffee futures returns tend to increase as temperatures approach freezing. One possible explanation is that market participants begin to price in the risk of frost-related supply disruptions as temperatures fall. This is consistent with USDA reports noting that frost does not necessarily damage the coffee beans directly, particularly when harvesting has already taken place. Instead, frost may damage the coffee trees themselves, potentially reducing future yields and creating more persistent effects on production. Anticipation of these longer-term supply risks may therefore lead traders to bid up coffee futures

prices. The table also shows that the standard deviation of returns rises substantially as temperatures decline, indicating that price movements become more volatile when frost risk is elevated. This suggests greater uncertainty in the market as temperatures approach levels associated with potentially damaging frost events.

After finding this out we then look to see if frost can provide any valuable signals and if it affects coffee prices.

$$R_t = \alpha + \beta W_t + \varepsilon_t, \quad (5.5)$$

this equation is a simple linear equation used to test the relationship between the minimum temperature and the price of coffee futures. However we know that the effect of temperature is not actually linear and for example if the temperature were to drop by one degrees and go from 1 to 0 degrees the effects would be far more devastating than if the temperature were to drop from 10 to 7 degrees. thats why we define a new equation that can try to capture this relationship.

$$R_t = \alpha + \beta_1 \max(0, T^* - T_t^{\text{Brazil}}) + \beta_2 \left[\max(0, T^* - T_t^{\text{Brazil}}) \right]^2 + \varepsilon_t, \quad (5.6)$$

In this equation T^* is the threshold for frost. We choose 5 to be the threshold, simply because from our analysis and what the paper says, the threshold for the frost does not actually make a difference in terms of showing if frost produces out of the ordinary effects, but it allows us to use more data points because unlike within the paper [7] brazil does not have as many cold spells like Florida. The results from the regression can be seen in this table

Table 5.14: Regression results for Arabica coffee futures returns and minimum temperatures

Model	α	β	β_1	β_2	R^2 (%)	Adj. R^2 (%)	Joint p	N
Linear	0.0285 (0.13)	-0.0043 (-0.22)	-	-	0.00	-0.02	-	3850
Quadratic	-0.0120 (-0.32)	-	-1.1650 (-2.87)	0.3175 (2.60)	1.53	1.48	0.0162	3850

We measure Brazilian frost risk using daily minimum 2-metre air temperature (T2MIN) from nasa power as we have been using, for the three Arabica-growing states: Minas Gerais, São Paulo, and Paraná, over January 1990 to July 2026. Because a damaging frost in any one region constitutes a supply shock, our temperature variable T^* is the minimum across the three regions on each day. To align the information sets, each trading day's return is matched with the minimum temperature realized over the interval since the previous close: for a normal Tuesday, this is Tuesday's overnight minimum; for a Monday, it is the minimum over Saturday through Monday, since a weekend frost is first priced in Monday's session. We restrict the sample to the southern hemisphere winter season (May–September), which yields 3,850 trading-day observations; frost is essentially absent outside this window.

The linear model has virtually no explanatory power. The estimated temperature coefficient

is negligible ($\beta = -0.0043$), with a t -statistic of -0.22 , while the model explains none of the variation in daily returns ($R^2 = 0.00\%$). On the basis of this specification alone, one would conclude that Brazilian temperatures bear no systematic relationship to Arabica coffee futures prices.

The piecewise quadratic specification produces a different result. Both freeze coefficients are statistically significant ($\beta_1 = -1.17$, $t = -2.87$; $\beta_2 = 0.32$, $t = 2.60$), and the joint hypothesis that $\beta_1 = \beta_2 = 0$ is rejected ($p = 0.016$). Although the resulting R^2 remains modest at 1.53%, this represents a substantial improvement for a daily asset-return regression, particularly given that only 162 of the 3,850 trading days in the sample contain a frost signal.

The economic intuition is reflected in the shape of the fitted response function. Because the positive quadratic coefficient dominates as temperatures decline, the relationship between returns and temperature is strongly convex. At the lowest observed temperatures, the model predicts daily returns approaching 10%, closely matching the realised price movements during the most severe frost episodes, including the 23.9% increase on 27 June 1994 and the 10.6% increase on 17 July 2000.

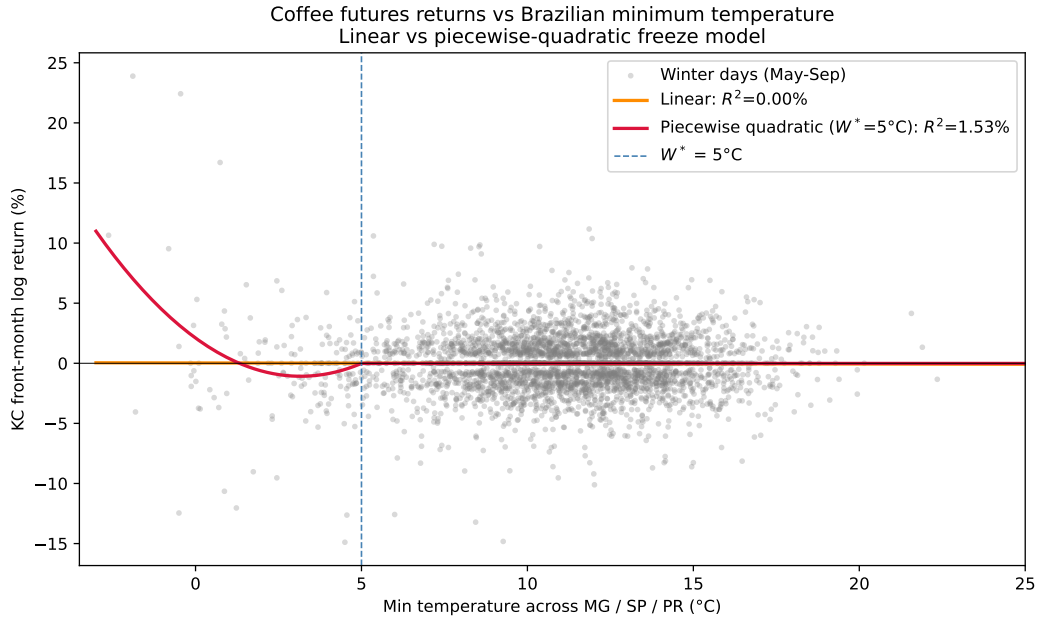


Figure 5.4: Estimated piecewise quadratic relationship.

The negative value of β_1 implies a small negative return for temperatures immediately below the threshold. Rather than indicating model inaccuracy, it is consistent with an anticipation effect. Cold weather is partially predictable, allowing market participants to incorporate some probability of frost into prices before the realised temperature is observed. Consequently, when temperatures fall close to—but do not exceed—the damage threshold, part

of the previously priced frost premium unwinds, producing a modest negative return. This represents a mild departure from the assumption that realised temperature is entirely unexpected, and the effect is expected to diminish as the threshold temperature, T^* , approaches the true biological damage point.

Taken together these findings, the relationship between futures prices and weather fundamentals is inherently nonlinear: prices respond primarily in states of the world where frost poses a genuine threat to future supply. Linear specifications, by averaging across both frost and non-frost regimes, fail to capture this state dependence and would therefore incorrectly conclude that coffee futures prices are largely unrelated to their underlying fundamentals.

These findings mirror, in Brazilian Arabica, the pattern established in the frozen orange juice literature. Roll [44] found that temperature explained only one to two per cent of FCOJ return variation, Boudoukh et al [7] claimed instead that actually a non linear model could potentially explain more of the variation, since a crop that is unaffected above freezing and severely damaged below it exhibits a threshold response that a linear regression averages away. The same mechanism operates here. Fitted linearly, minimum temperature explains none of the variation in coffee returns, with an R^2 of 0.00 %, a result that taken alone would suggest Arabica prices are unrelated to Brazilian weather. Permitting the response to break at $T^* = 5^\circ\text{C}$ recovers an R^2 of 1.53 %, with the entire improvement arising below the breakpoint while the relationship remains flat across the rest of the temperature range. The economic relationship is therefore present but confined to the frost regime, and the appropriate way to condition on temperature is through a threshold rather than a continuous variable, a conclusion that motivates the trading rules developed in Chapter 7.

Chapter 6

Why a strategy rather than a derivative

6.1 Challenges in Constructing a Climate Derivative

The original plan following the brief for this project was to create a derivative product which allowed the expression of ones views on climate through the buying and selling of the product, however as we will see in this section it may actually be better to try and create a quantitative investment strategy and offer that instead to the client for reasons that will be discussed. We will then use the findings from the previous chapter in order to create and back-test a strategy based on the frost affect on coffee in the following chapter.

6.1.1 Defining an Appropriate Underlying

The first challenge is identifying an underlying variable that captures the climate exposure of interest. Climate change affects financial markets through several channels, including physical risk, transition risk and liability risk. However, these risks are not themselves traded and it may not be correct to proxy them with something. If we were to say that the derivative can be written on a physical climate variable such as temperature or rainfall then the difficulty is that the economic effect of such a variable depends on both location and exposure. A temperature anomaly in one region may have a significant impact on agricultural production while having little direct financial relevance elsewhere. Consequently, a derivative written on a broad climate index may provide an imperfect hedge for the specific economic exposure faced by an investor. Conversely, defining a highly specific climate index may improve the economic relationship between the underlying and the exposure being hedged, but at the cost of reducing the number of natural buyers and sellers of the resulting contract. There is therefore an inherent trade-off between the specificity of the climate exposure and the potential depth of the market.

6.1.2 Basis Risk

Even with a well-chosen underlying, the relationship between X_t and S_t is not deterministic. Write

$$\Delta S_t = \beta \Delta X_t + \varepsilon_t, \quad R^2 = \frac{\beta^2 \text{Var}(\Delta X_t)}{\text{Var}(\Delta S_t)} \quad (6.1)$$

where ε_t captures everything else driving the asset price — inventories, FX, energy costs, geopolitics, expectations about future supply. This is basis risk: a derivative paying $f(X_T)$ hedges only the $\beta \Delta X_t$ component, leaving ε_t unhedged. This is consistent with, not contradicted by, the predictive results in Chapter 5. The Faccini/Ma et al. regressions find PC1 significant at all horizons with R^2 up to 6.32% at twelve months, and the frost-threshold model recovers a further 1.53%. Both are genuine evidence that climate carries exploitable information but even the strongest of these figures leaves over 93% of $\text{Var}(\Delta S_t)$ unexplained. A small, statistically robust R^2 is enough to build a systematic tilt into a strategy; it is not enough for X_t to serve as a reliable derivative underlying. Section 6.3 develops this distinction.

6.1.3 Liquidity and Price Discovery

A derivative market requires participants willing to take both sides of a transaction. This is particularly problematic when the underlying climate exposure is highly specialised. Established futures markets benefit from standardisation and the concentration of trading activity into a relatively small number of contracts. A bespoke climate derivative instead risks fragmenting liquidity across locations, maturities and climate variables. Limited trading activity would result in wider bid-ask spreads and weaker price discovery, increasing the cost of entering and exiting positions.

Liquidity is also important for valuation. In a sufficiently liquid market, traded prices provide information that can be used to calibrate pricing models. Where the underlying climate risk is not actively traded, the model must rely more heavily on assumptions about the physical probability distribution and the market price of climate risk. Valuation therefore becomes both more model-dependent and more difficult to verify empirically.

6.1.4 Hedging and Warehousing Risk

A further challenge arises from the position of the derivative seller. For any derivative contract to exist, one party must take the opposite side of the buyer's exposure. If the derivative has terminal payoff X_T , then the seller is effectively exposed to $-X_T$. In conventional derivatives, this risk can often be reduced by constructing a hedge H_T using the underlying asset or a closely related traded instrument, with the aim that $H_T \approx X_T$. However, for derivatives written on non-traded climate variables such as temperature or rainfall, no directly tradable underlying asset may exist. As a result, the hedge may be imperfect, leaving residual exposure $X_T - H_T \neq 0$, which the seller must retain. This problem becomes more severe for long-dated contracts, since available hedging instruments may have much shorter maturities than the derivative itself. The seller must therefore repeatedly replace

or roll the hedge, creating both basis risk, due to the imperfect relationship between the hedge and the climate exposure, and rollover risk, due to uncertainty over the future cost and availability of replacement hedges.

6.1.5 Counterparty, Margin and Funding Considerations

Long-dated derivative positions also introduce counterparty and funding considerations. Let V_t denote the mark-to-market value of the position and M the collateral or maintenance margin posted against it. A margin call is triggered whenever the interim loss exceeds the posted margin, that is whenever

$$V_t < -M \quad (6.2)$$

for some $t < T$, regardless of the terminal payoff V_T . This means profitability at maturity is not sufficient for the position to be feasible: what matters is

$$\inf_{t \leq T} V_t > -M, \quad (6.3)$$

a path-wise condition rather than a terminal one. Even if an investor's long-run climate view is ultimately correct, in the sense that $\mathbb{E}[V_T] > 0$, a large adverse move at some intermediate t can force the position to be unwound before it has the chance to pay off, if the investor or counterparty cannot finance the interim loss.

6.1.6 Pricing and the Market Price of Climate Risk

The pricing of a climate derivative presents an additional conceptual difficulty. In conventional derivative pricing, a sufficiently complete set of traded instruments allows the payoff $f(X_T)$ to be replicated by a self-financing portfolio in the underlying, which is what permits risk-neutral valuation

$$P_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} f(X_T) \mid \mathcal{F}_t \right] \quad (6.4)$$

under a unique pricing measure $\mathbb{Q}$. A climate variable such as temperature or rainfall is not itself a traded asset, so a contingent claim written on it is generally not perfectly replicable: there is no portfolio in traded instruments whose value tracks $f(X_T)$ path-by-path. The market is incomplete, and rather than a single $\mathbb{Q}$ there is instead a whole family of measures $\{\mathbb{Q}_i\}_{i \in I}$ consistent with the traded prices observed, each assigning a different value to the claim.

If valuation instead relies on the physical probability distribution, this amounts to choosing a market price of climate risk λ , which shifts the drift of X_t when moving from the physical measure $\mathbb{P}$ to a pricing measure: if X_t evolves as $dX_t = \mu dt + \sigma dW_t$ under $\mathbb{P}$, the corresponding pricing dynamics are

$$dX_t = (\mu - \lambda\sigma) dt + \sigma dW_t. \quad (6.5)$$

Estimating λ is challenging precisely because liquid instruments directly exposing investors to X_t may not exist: λ is ordinarily inferred from the prices of traded assets that already carry the risk in question, and no such assets are available here. Consequently, the price P_t

can become highly sensitive to the assumed value of λ , or equivalently to the choice of θ , with no traded price available to discipline either. This limits both transparency and the ability to validate the proposed valuation framework using observed market prices.

6.2 A Quantitative Investment Strategy as an Alternative

The difficulties described above motivate a different approach. Rather than creating a new derivative whose payoff depends directly on a climate variable, this dissertation constructs a Quantitative Investment Strategy using existing traded instruments.

The central idea is to shift the problem from

“How should a new climate contingent claim be priced and hedged?”

to

“Can climate-related information be mapped systematically into positions in existing liquid markets?”

This distinction is important. The QIS does not eliminate climate risk or the modelling uncertainty associated with it. Instead, it changes the implementation mechanism through which an investor obtains climate-related financial exposure.

6.2.1 Use of Existing Liquid Markets

The principal advantage of the QIS approach is that positions can be implemented through instruments that already trade in established markets. In the context of this dissertation, commodity futures provide a particularly natural implementation mechanism because the prices of agricultural commodities can be sensitive to weather and other physical climate risks. Using established futures contracts avoids the need to create liquidity in a new climate derivative. Market prices, transaction mechanisms, margining conventions and contract specifications already exist.

6.2.2 Transparency and Replicability

A systematic strategy can be specified through a deterministic set of rules: the position at each date is given by

$$w_t = g(\mathcal{F}_t; \theta) \tag{6.6}$$

for some fixed, publicly specified function g and parameter vector θ known at inception. At each rebalancing date t , this rule-based structure has an important advantage: because g and θ are fixed and $\mathcal{F}_t$ is observable historical data, anyone with access to $\{X_t, S_t\}_{t \leq T}$ can recompute w_t exactly and reconstruct the strategy’s entire historical path independently. The investment process is therefore more transparent than a bespoke derivative, whose valuation P_t depends on the market price of climate risk λ — a parameter that is not observed, cannot be recomputed from public data, and is instead assumed by whoever is pricing the contract.

6.2.3 Continuous Rebalancing of the Climate View

A derivative typically fixes its payoff function $f(\cdot)$ at inception $t = 0$, so the terminal payoff $f(X_T)$ is determined once and for all by information available at that single date. A QIS can instead update its positions as new information becomes available: the strategy holds a position $w_t = w(\mathcal{F}_t)$ at each date t , where $\mathcal{F}_t$ is the information set available at t , and w_t is free to change as $\mathcal{F}_t$ evolves. This flexibility is particularly useful for climate positioning because the financial implications of climate change evolve through time. New weather observations, changes in expectations, shifts in inventories, adaptation, and changes in futures curves all update $\mathcal{F}_t$ and can alter the attractiveness of a position. A systematic strategy can incorporate these changes through periodic rebalancing, so that its cumulative payoff

$$\sum_{t=0}^{T-1} \theta_t (S_{t+1} - S_t) \quad (6.7)$$

depends on the entire sequence of positions taken along the path, rather than on a single function of the terminal state $f(X_T)$. Climate positioning therefore becomes dynamic rather than being restricted to a single terminal payoff.

6.2.4 Scalability

The capacity of a QIS is ultimately constrained by the liquidity of its constituent instruments. If the strategy trades established futures markets, this capacity can be evaluated directly using observed trading volumes and open interest. By contrast, the capacity of a newly created climate derivative is initially unknown because the market itself does not yet exist. A QIS therefore provides a clearer route to evaluating whether the proposed climate exposure could be implemented at commercially meaningful scale.

6.2.5 Observable Performance and Backtesting

A further advantage is that a QIS produces a time series of returns that can be evaluated empirically. If $R_{i,t+1}$ denotes the return of instrument i between t and $t + 1$, the strategy return can be written as

$$R_{t+1}^{\text{QIS}} = \sum_{i=1}^N w_{i,t} R_{i,t+1} - C_{t+1}, \quad (6.8)$$

where C_{t+1} represents transaction and implementation costs. The historical strategy can therefore be assessed using conventional measures of investment performance and risk, including annualised return, volatility, Sharpe ratio, drawdown and turnover. More importantly, the source of the strategy's returns can be investigated directly. For a climate-positioning strategy, strong historical performance alone is insufficient. The analysis must establish whether returns are plausibly associated with the climate mechanism the strategy is intended to capture rather than unrelated risk premia or accidental correlations.

6.3 What the QIS Does Not Solve

The move from a bespoke derivative to a QIS should not be interpreted as eliminating the underlying economic difficulties. First, basis risk remains. The traded instruments used by the strategy are imperfect proxies for climate risk and remain exposed to many non-climate determinants. Second, the strategy introduces its own implementation risks. Futures positions require margin, contracts must be rolled, and transaction costs may reduce returns. Futures-curve dynamics can also generate significant positive or negative roll returns. Third, historical backtesting introduces model-selection and overfitting risk. A strategy constructed using historical climate and market data may appear successful because its parameters have inadvertently been selected to fit the sample. The methodology must therefore distinguish clearly between model development and out-of-sample evaluation. Finally, a QIS does not provide the same payoff certainty as a derivative. A derivative can be designed to generate a specific payoff conditional on a predefined event. A QIS instead generates a path-dependent investment return determined by both the trading signal and subsequent market movements.

The QIS should therefore be regarded as an alternative implementation of climate positioning rather than a direct substitute for every possible climate derivative. Nonetheless, the remainder of this dissertation develops this approach. The objective is to determine whether climate-related information can be translated into a transparent set of trading rules, implemented using sufficiently liquid instruments, and whether the resulting strategy exhibits economically meaningful performance after accounting for the principal costs and risks of implementation.

Chapter 7

Creating and Back-testing strategies

In this chapter we look to create and back-test trading strategies that involve taking a position based on a belief we have about climate. This will allow us to offset some of the risk induced by climate related risks in the coffee markets. We have established in chapter 5 that frost has an affect on coffee production due to the damage it does to crops and future yields which of course has an affect on the price. In order to better understand the reaction of the market to frost let us view a plot of the price with major frost events marked.

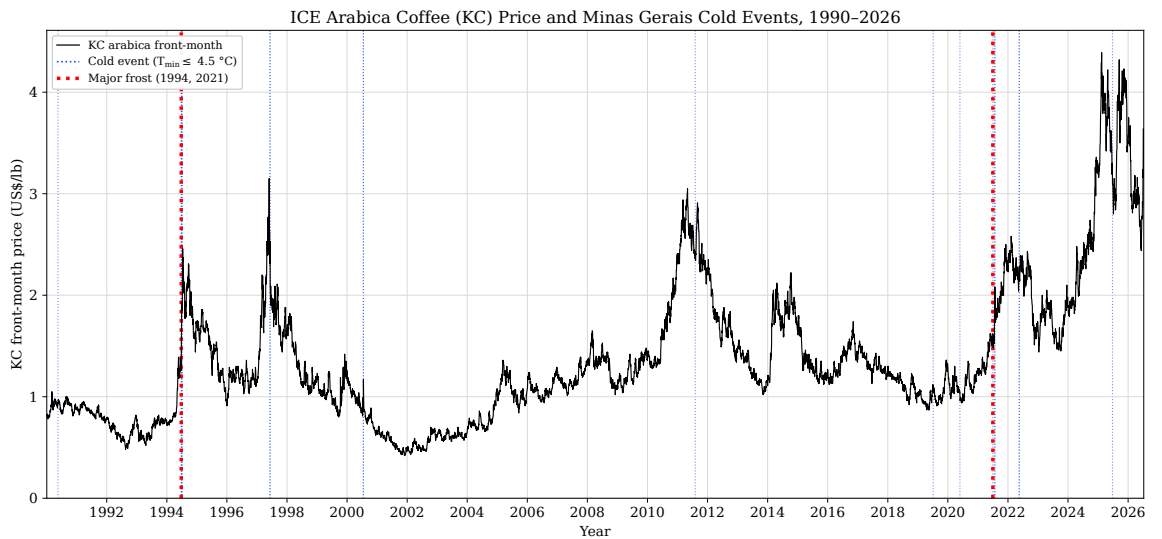


Figure 7.1: Plot of price with major frost events and $T_{\min} \leq 4.5^{\circ}\text{C}$

From 7.1 we can see that some of the big moves that occurred in this chart were very close to major frost events or major indicators. The two red lines show the periods when frost damage were confirmed in many parts of Brazil which as visible in the chart resulted in substantial hikes in price for coffee. The 1994 Brazilian frost had a substantial impact on Coffee C futures prices. Immediately prior to the first severe frost on 25–26 June,

Coffee C was trading at approximately \$1.26 per pound. Following the frost, prices rose to approximately \$1.60 per pound on the next trading day, representing an increase of around 27%. A second severe frost occurred on 9–10 July, further increasing concerns regarding damage to Brazilian coffee production and contributing to a subsequent peak of approximately \$2.46 per pound. Consequently, between the price immediately preceding the first frost and the subsequent peak, Coffee C increased by approximately 95% in less than three weeks.

The behaviour of coffee prices during 1997 provides an important example of how expectations of adverse weather can influence futures prices even in the absence of a realised frost event. Coffee prices had already risen substantially during the early months of 1997 due to tight global inventories and concerns over limited supplies. However, as the Brazilian winter approached, increasing fears that cold weather could produce another damaging frost contributed to a sharp acceleration in prices. In late May, Coffee C futures rose to approximately \$3.00 per pound, reaching their highest level in around twenty years, as traders responded to forecasts of colder weather and the possibility of frost in Brazil's major coffee-producing regions. Importantly, a damaging frost did not subsequently materialise. As weather forecasts improved and indicated that temperatures in the principal coffee-growing regions were likely to remain above approximately 6°C, frost concerns diminished and coffee futures fell sharply. The 1997 episode therefore illustrates that coffee prices can respond strongly not only to realised frost damage, but also to changes in the market's expectations of future frost risk.

The 2021 Brazilian frost also had a substantial impact on Coffee C futures prices. A severe frost affected Brazil's coffee-growing regions on 20 July, with particularly significant damage reported across important Arabica-producing areas, including southern Minas Gerais. On 20 July, New York Arabica futures were trading at approximately \$1.66 per pound. As the extent of the frost damage became apparent and concerns regarding future Brazilian coffee production intensified, prices increased rapidly, reaching approximately \$2.08 per pound by 26 July. This represented an increase of approximately 25.4% in less than one week. The September Arabica futures contract reached an intraday high of approximately \$2.15 per pound on 26 July, its highest level since 2014. The rapid price appreciation following the frost events demonstrates the sensitivity of Coffee C futures to severe weather conditions in Brazil and, in particular, to changes in expectations regarding future Brazilian Arabica supply.

Frost can have both short- and long-term effects on production, depending on its severity. Limited damage to the current harvest may primarily increase prices of near-term contracts, whereas severe damage to coffee trees can reduce production for several years and therefore affect longer-dated contracts. Figure 7.2 illustrates this by comparing contracts across different maturities, while also showing whether the market is in contango or backwardation. This is particularly evident following the 2021 “black frost”, which damaged coffee trees and coincided with a shift from a prolonged period of contango to multi-year backwardation, demonstrating how severe frost events can alter both current prices and the structure of

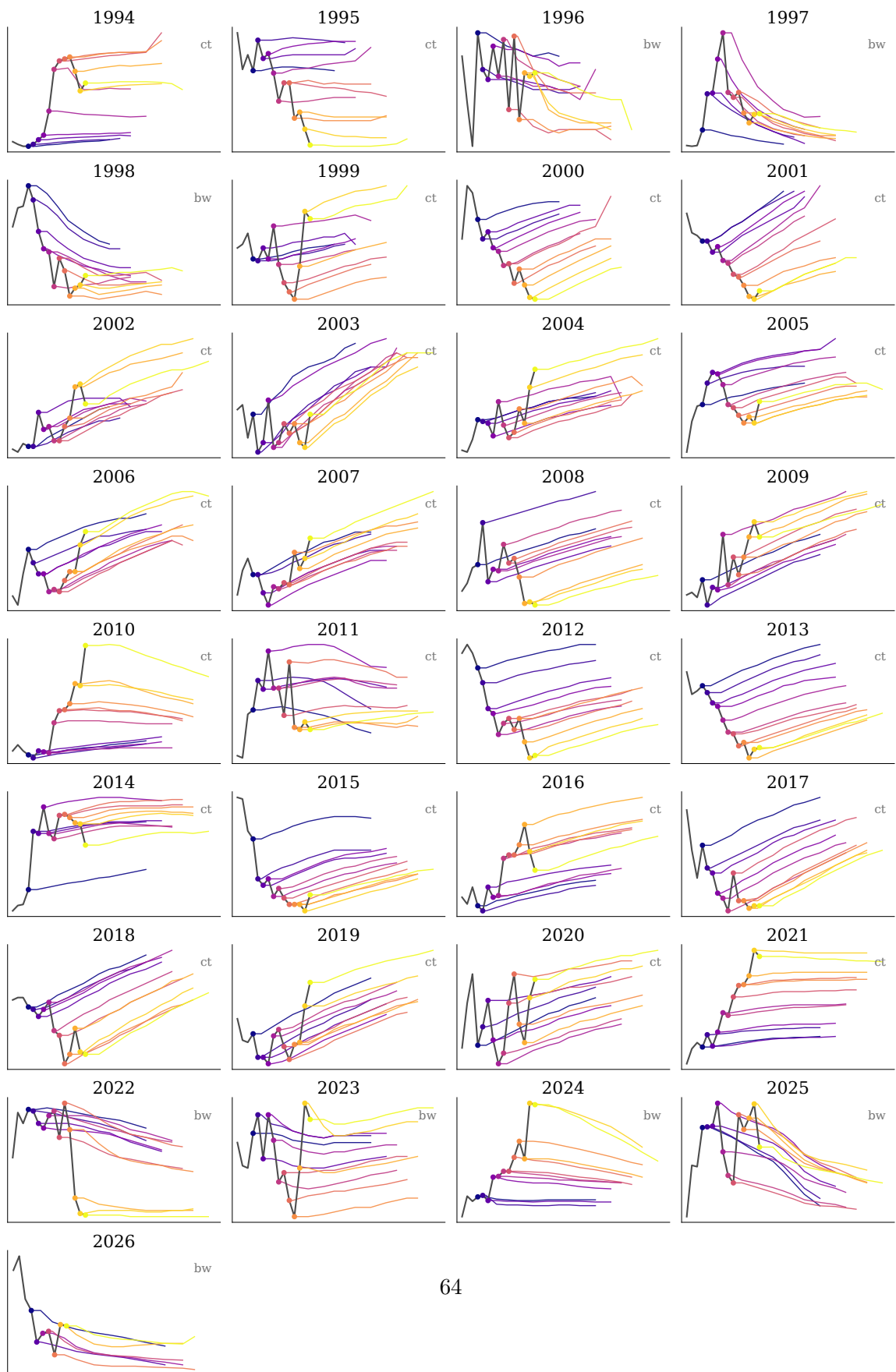


Figure 7.2: Monthly futures C1 price plus 2 year forward, Arabica. ct is contango and bw is backwardation

the futures curve. A closer look at what happens around the frost events is shown in 7.3, which just shows the things discussed below 7.1 in visual form.

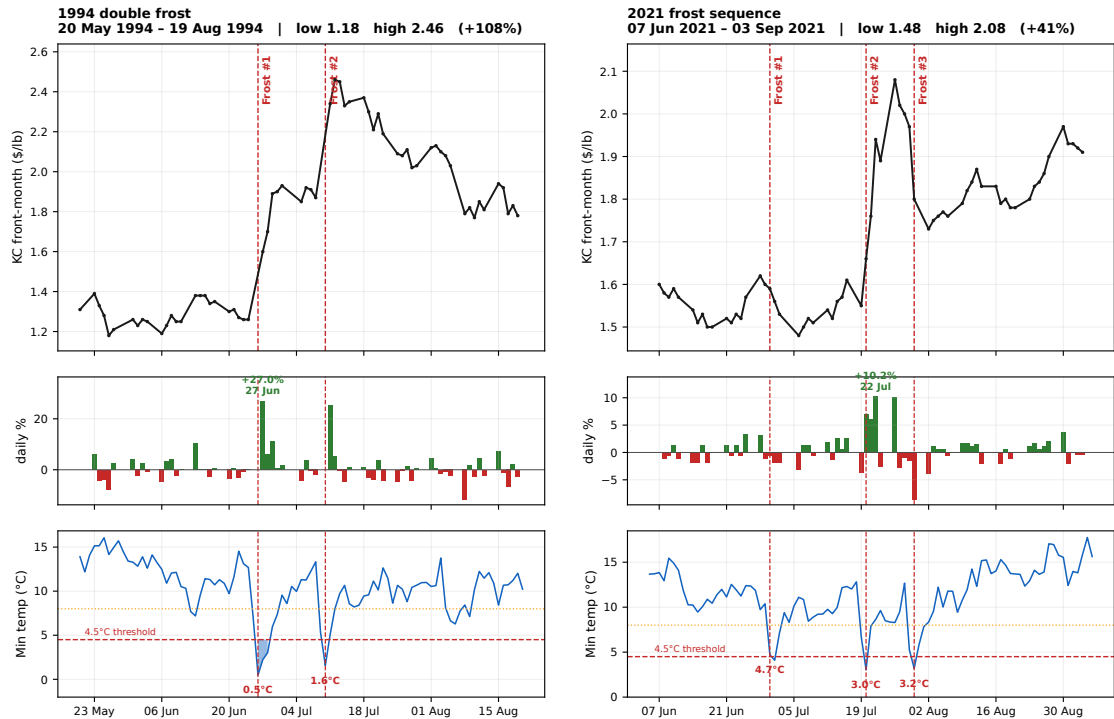


Figure 7.3: Price of coffee near big frost events zoomed in.

7.1 1st strategy

The first strategy translates a weather observation directly into a trading position. When the daily minimum temperature in Brazil's main coffee growing region (Minas Gerais) falls below a frost threshold (in this case 4.5°C), it buys arabica coffee futures and holds them for ten trading days, then closes the position regardless of outcome. At all other times it holds no position. The economic reasoning is straightforward. Brazil produces roughly a third of the world's arabica as we have investigated in chapter 5, and Minas Gerais alone accounts for about half of Brazilian output. Arabica has almost no cold tolerance: leaf tissue is damaged near 0°C , and a severe frost kills the branch wood that would have carried the following season's crop. Because replacing damaged trees takes years, supply is highly inelastic over a one-year horizon, and a genuine frost forces a large upward price adjustment. The trade rests on the assumption that this adjustment is not instantaneous. When cold weather occurs, nobody yet knows how much damage has been done, frost burnt leaves and dead branches look identical for the first fortnight, and assessing the true extent requires field surveys and revised crop forecasts. If prices adjust gradually as this information arrives, a position opened after the cold event and held through the assessment window should earn a positive return.

7.1.1 How it was constructed

Temperature comes from NASA POWER (MERRA-2 reanalysis) for the grid cell at -21.55° latitude, -45.43° longitude, in the Sul de Minas producing region, daily from 1990 to 2026. The variable used is the daily minimum air temperature at 2 m. Prices are ICE Arabica Coffee “C” front-month daily settlements over the same period obtained from LSEG (like in the other chapters in this dissertation). On a side note the reason we use only Minas Gerais unlike the multitude of Brazilian locations we had in chapter 5 is because Minas Gerais as mentioned is the biggest region as well as the fact that due to Parana being very frost prone a lot of the coffee production moved from there after 1975. Sao Paolo is also not very frost prone anyway hence we did not include it in the signal

A signal is generated when the minimum temperature falls below 4.5°C . Consecutive cold days do not re-trigger; only the first breach counts as a new event. This threshold is a proxy rather than a physical frost line. MERRA-2 output is a grid-cell average over an area of roughly 50 km by 60 km at 869 m mean elevation, which smooths away the localised frost that forms in low-lying valleys on clear, still nights. A grid mean of 4.5°C is consistent with sub-zero temperatures in the coldest pockets of that cell, the July 2021 event recorded 3.0°C on the grid while ground stations in Minas Gerais reported temperatures as low as -1.2°C .

Temperature is recorded every calendar day, but futures trade only on business days, so a frost falling on a weekend risks being lost. Each trading session is therefore assigned the coldest minimum observed across all days since the previous session closed. Without this adjustment three of the eleven events in the sample disappear.

Entry is at the close of the session after the signal. This matters: a daily minimum is only knowable once the day has ended, so entering at the signal day’s own close would use information unavailable at the time and constitute look-ahead bias. The position is held for ten trading days and exited at the close. New signals arriving while a position is open are ignored, so trades never overlap. Transaction costs of 5 basis points per side are applied.

7.1.2 Results

Table 7.1: Performance of the 10-day frost strategy, 1990–2026.

Metric	Value
Total return (%)	−2.50
CAGR (%)	−0.07
Annualised volatility (%)	7.22
Sharpe ratio	0.025
Sortino ratio	0.004
Maximum drawdown (%)	−34.42
Calmar ratio	−0.002
Time in market (%)	1.20

Table 7.2: Trade-level statistics.

Statistic	Value
Trades	11
Hit rate	36 %
Mean trade	+1.02%
Median trade	−3.13%
Best trade	+44.0% (June 1994)
Worst trade	−26.6% (July 2000)
<i>t</i> -statistic vs zero	0.19

The Sharpe ratio is not informative here. Because the strategy holds risk on only 1.2 % of sessions, the daily return series is almost entirely zeros, which deflates the standard deviation and makes the annualised ratio close to meaningless. The Sharpe measured only over days when a position is actually held is 0.25, better than flat, but still far from a tradeable edge.

The mean return depends entirely on one observation. The average trade is positive only because of June 1994, which returned +44% by capturing the genuine double frost that cut the following Brazilian crop roughly in half. Excluding that single trade, the mean falls to −3.3%. The median trade is negative, and a *t*-statistic of 0.19 on eleven observations cannot distinguish the strategy from noise at any conventional level of significance.

Figure 7.4 makes the character of the result visible: long flat stretches in cash, punctuated by a handful of steps. The gain from 1994 is fully surrendered by the July 2000 trade, after which the strategy never recovers its earlier peak, leaving a maximum drawdown of 34 % that persists for most of the remaining sample.

Validated against the documented record, the detector fires on 26 June and 10 July 1994, the two dates of the recognised “double frost” and on 30 June, 20 July and 30 July 2021, matching exactly the three severe frost instances recorded by the USDA Foreign Agricultural Service. The filter recovers both major events of the sample period. Its weakness is precision rather than recall: it also fires in years with no documented crop damage, most notably July 2000, when the resulting position lost 27 %.

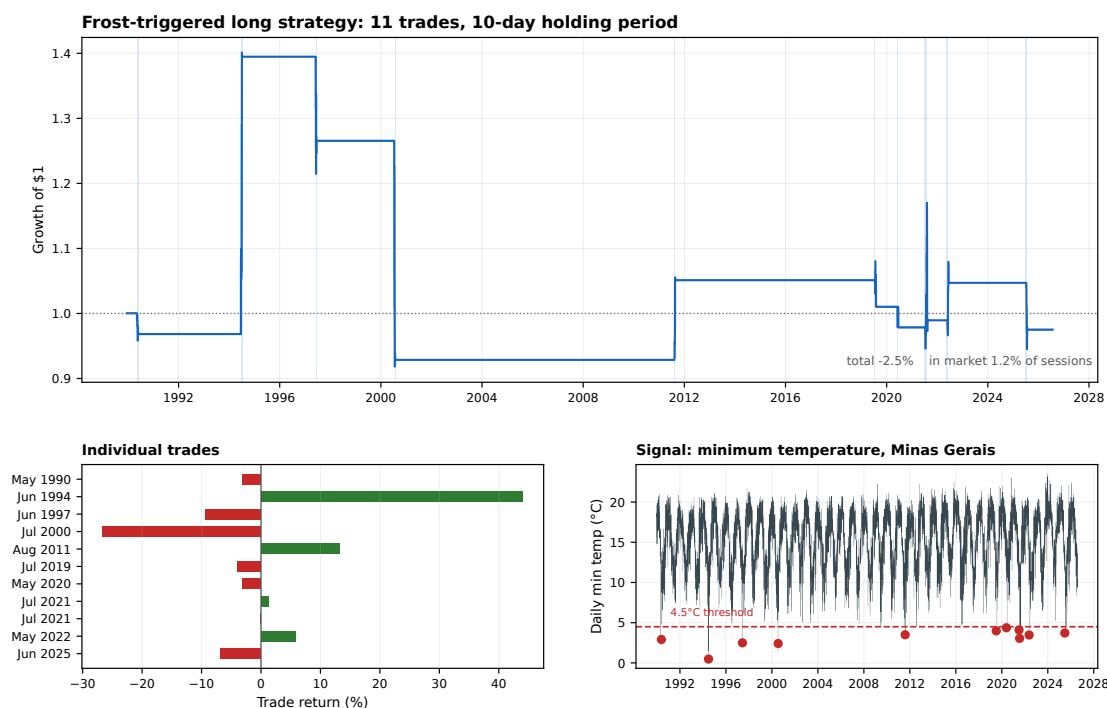


Figure 7.4: Equity curve, trade history, and signals for strategy 1. Shaded bands are where strategy is long

7.2 2nd strategy

The strategy converts the same weather observation into a substantially longer position. When the daily minimum temperature in Minas Gerais falls below 4.5 °C, it buys arabica coffee futures at the following session's close and holds them for sixty trading days before exiting, irrespective of what has happened in the interim. At all other times it holds no position. Only the exit rule changes. This isolates the holding period as the single variable under test, so any difference in outcome is attributable to how long the position is held rather than to a different view of when frost has occurred.

The rationale for lengthening the horizon follows from these reasons, severe frost damages branch wood, so the supply shock impairs the *following* season's crop rather than the current one. Establishing the extent of that impairment is slow: frost-burnt leaves and dead branches are visually indistinguishable for roughly the first fortnight, and the true loss emerges only as agronomists survey affected farms, cooperatives aggregate reports, and agencies such as CONAB revise official crop forecasts. Ten trading days is shorter than this assessment cycle. A three-month window is matched to the horizon over which the fundamental information is actually revealed, and is chosen on that basis rather than selected to maximise returns.

7.2.1 How it was constructed

The data and signal are both unchanged from the previous section. Temperature is NASA POWER (MERRA-2 reanalysis) for the grid cell at -21.55° latitude, -45.43° longitude; prices are ICE Arabica Coffee “C” front-month daily settlements. A signal is generated on the first breach of 4.5°C , with consecutive cold days not re-triggering, and each trading session is assigned the coldest minimum observed since the previous session closed.

Entry remains at the close of the session *after* the signal, avoiding look-ahead bias. The position is held for sixty trading days and exited at the close. New signals arriving while a position is open are ignored. Transaction costs of 5 basis points per side are applied.

Because each position now occupies three months rather than two weeks, more subsequent signals fall inside an open trade and are absorbed by the non-overlap rule. The same 18 signal days therefore collapse to 10 non-overlapping trades rather than eleven. Notably, the second frost of July 1994 is absorbed into the position opened after the first, so the two are captured as a single trade rather than two.

7.2.2 Results

Table 7.3: Performance of the 60-day frost strategy, 1990–2026.

Metric	Value
Total return (%)	49.50
CAGR (%)	1.11
Annualised volatility (%)	11.75
Sharpe ratio	0.152
Sortino ratio	0.055
Maximum drawdown (%)	−43.05
Calmar ratio	0.026
Time in market (%)	6.55

Table 7.4: Trade-level statistics across the ten signals.

Statistic	Value
Trades	10
Hit rate	40 %
Mean trade	+5.80%
Median trade	−1.65%
Average win	+25.8%
Average loss	−7.6%
Win–loss ratio	3.41
Best trade	+34.0% (June 1994)
Worst trade	−29.2% (July 2000)
Skewness	−0.09
<i>t</i> -statistic vs zero	0.94

The strategy is profitable, returning 49.5 % over thirty-six years against a loss of 2.5 % at the ten-day horizon. Since the signal is identical, the entire difference is attributable to the exit rule. The position is held on only 6.6 % of trading sessions, so the strategy is in cash for the overwhelming majority of the sample.

The payoff is asymmetric. The strategy wins on only four of ten trades yet earns a positive mean of 5.80 %, because the average win of 25.8 % is more than three times the average loss of -7.6 %. A win-loss ratio of 3.41 is what makes a 40 % hit rate profitable, and the median trade of -1.65 % confirms that the result is driven by the right tail rather than by typical performance. This follows from the agronomy: when frost occurs it destroys productive capacity for several seasons and prices must rise sharply to ration demand, as in June 1994 (+34.0%) and July 2021 (+30.0%); when it does not, the position merely carries ordinary market volatility for three months.

Realised skewness of -0.09 is essentially zero rather than positive, because the worst trade (July 2000, -29.2%) is comparable in magnitude to the best. Long futures provide linear exposure in both directions, so a three-month holding period extends participation in adverse moves exactly as fully as in favourable ones. A convex payoff of the kind the insurance analogy implies would require an instrument with limited downside, such as a call option, which futures cannot replicate.

A t -statistic of 0.94 on ten observations does not approach conventional significance. The strategy is directionally consistent with the hypothesis but cannot be distinguished from a random long position of equivalent duration. The Sortino ratio of 0.055, well below the Sharpe of 0.152, further indicates that losing periods are more volatile than winning ones.

7.2.3 Robustness

Table 7.5: Total return across the threshold-horizon grid.

Threshold	20 days	40 days	60 days	90 days
3.5 °C	-41.4	-37.9	-8.7	-48.6
4.5 °C	-16.8	3.5	49.5	-18.6
5.5 °C	2.2	23.6	26.7	4.5
6.5 °C	-34.1	-25.7	-30.5	-38.6

The result is not robust, and this must be stated plainly: at the chosen threshold the sign of the return changes with the holding period, and the profitable region is confined to a band around 4.5 °C to 5.5 °C at 40 to 60 days. Two observations qualify this. The failure at 6.5 °C is economically coherent, since at that threshold the filter admits ordinary cool nights that pose no agronomic threat, so no supply shock follows and the position is simply a random long. The failure at 3.5 °C reflects sample exhaustion, with too few trades remaining for a stable estimate. Whether the intermediate band constitutes economic content or a fortunate coincidence cannot be resolved with ten observations.

Table 7.6: Sub-period stability.

Period	Total return (%)	Sharpe
1990–2001	−13.0	0.00
2002–2013	−4.8	−0.05
2014–2026	80.4	0.45

The entire profit accrues in the final third of the as seen in the equity curve in 7.5; the first two sub-periods are marginally negative. This is the most serious limitation of the result. An economic reading is available, in that global ending stocks have declined over the period and a market with thinner inventory buffers must reprice supply shocks more violently. A competing and less flattering reading is that the recent trades coincided with unrelated bullish drivers. The June 2025 trade illustrates the difficulty: it returned +18.3%, but the dominant driver of coffee prices over that window was the imposition of a 50 % United States tariff on Brazilian imports effective 1 August 2025, which disrupted trade flows and drove certified inventories to multi-year lows. The strategy conditions only on temperature, so any other shock affecting coffee during the holding window enters the measured return.

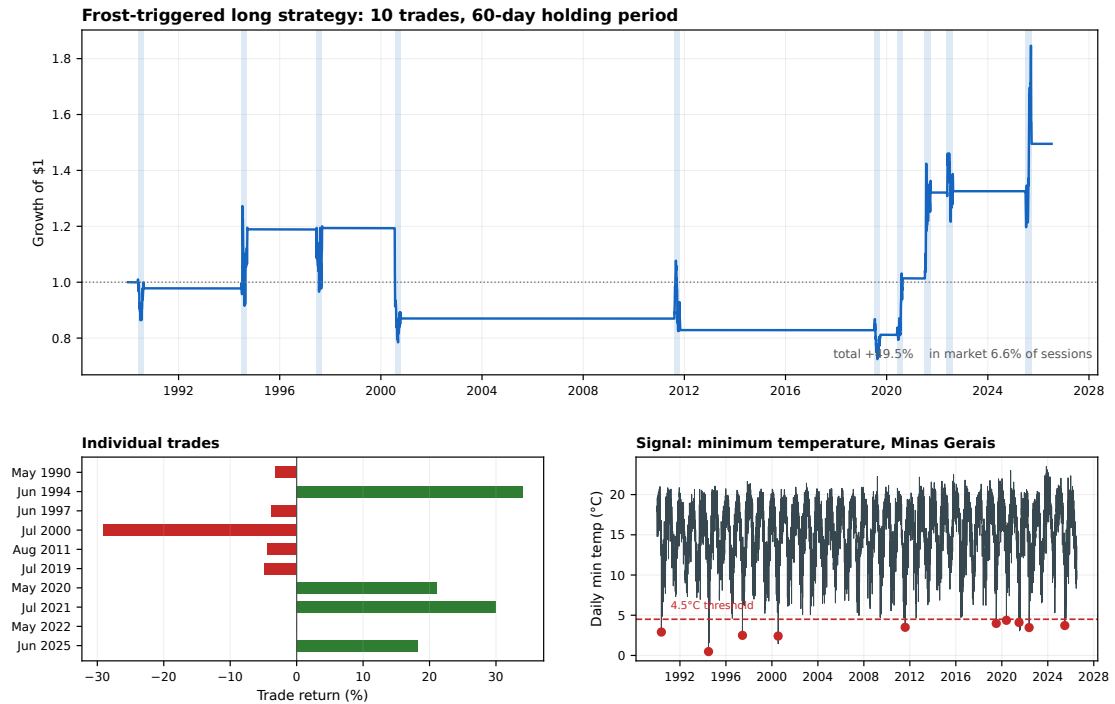


Figure 7.5: Equity curve, trade history, and signals for strategy 2. Shaded bands are where strategy is long

7.3 3rd strategy

The first two strategies both bought coffee after cold weather, and both were built on the assumption that a frost threshold identifies impending crop damage. A recurring observation across those results was that the filter fired frequently in years where no damage occurred. The third strategy takes that observation as its starting point and inverts the trade.

7.3.1 What the strategy does

When an extreme cold anomaly appears in the Brazilian coffee belt during the frost season, the strategy sells arabica coffee futures at the following session's close and holds the short position for twenty trading days. The reasoning proceeds from a simple asymmetry in what the market can know. When cold weather occurs, nobody yet knows whether trees have died, because damage assessment takes weeks. The market cannot price the outcome, so it prices the distribution of outcomes: if a cold snap implies, say, a one-in-four chance of losing a material fraction of next year's crop, the price rises immediately by roughly the product of probability and severity. Over the following weeks that uncertainty resolves. In the majority of cases no material damage is confirmed, the premium was priced for a loss that did not occur, and it decays out of the price.

A short position initiated after the anomaly therefore collects that decay. The economic interpretation is that the strategy provides insurance. Roasters hedging input costs and speculators buying frost protection are both paying to avoid holding the risk that Brazil's crop is destroyed; the seller of that protection earns the payment in exchange for absorbing an infrequent but severe loss. This is the same structure as writing catastrophe bonds or selling variance, and it generates a characteristic return distribution: frequent small gains financed by rare large losses.

7.3.2 How it was constructed

Three changes distinguish this strategy from the previous two. Each is motivated by the underlying economic and physical mechanism rather than by the back-test results.

First, the strategy uses temperature anomalies rather than fixed thresholds. A minimum temperature of 6 °C in May may represent an unusually cold event, whereas the same temperature in July may be relatively normal. To account for this seasonality, a day-of-year climatology of minimum temperature is constructed and smoothed using a ± 7 -day window. The signal is then based on the deviation from this seasonal average. A five-day mean anomaly is used to reduce the effect of isolated noisy observations and to better capture the multi-day cold spells that are more relevant for frost risk. Since the anomaly is measured relative to normal conditions, signal days do not necessarily correspond to temperatures close to freezing. A signal is generated when the five-day anomaly crosses into the coldest 5 % of its historical distribution. This produces 67 signal events, compared with only eleven under the earlier 4.5 °C threshold, giving a much larger sample on which to evaluate the strategy.

Second, signals are only considered between 1 May and 1 September, which corresponds to the period in which frost risk is relevant for Brazilian coffee production. This restriction is therefore based on the physical mechanism rather than on strategy performance. When the seasonal restriction is removed, the Sharpe ratio falls from 0.22 to 0.06, which is consistent with the effect being concentrated in the period when frost risk is economically relevant.

Finally, positions are entered at the close of the trading session following the signal to avoid look-ahead bias. The strategy then holds a short position for twenty trading days, with transaction costs of 5 basis points applied on both entry and exit.

7.3.3 Results

Table 7.7: Performance of the frost premium strategy, 1990–2026.

Metric	Value
Total return (%)	118.51
CAGR (%)	2.17
Annualised volatility (%)	15.36
Sharpe ratio	0.218
Sortino ratio	0.101
Calmar ratio	0.047
Maximum drawdown (%)	−46.44
Time in market (%)	12.78

Table 7.8: Trade-level statistics across the fifty short trades.

Statistic	Value
Trades	50
Hit rate	60 %
Mean trade	+2.5%
Median trade	+3.34%
Average win	+9.5%
Average loss	−9.3%
Best trade	+30.6% (June 1997)
Worst trade	−46% (August 2025)
Skewness	−1.53
<i>t</i> -statistic vs zero	1.14

The strategy returns 118.5 % over thirty-six years while holding a position on 12.8 % of trading sessions. It is the only one of the three specifications to generate a materially positive return, and the only one with enough observations to permit meaningful inference. The strategy wins on 60 % of trades, and the median trade of 3.34 % exceeds the mean of 2.03 %. A median above the mean is the defining property of a left-skewed distribution: typical performance is better than average performance because a small number of severe losses drag the average down. Realised skewness of −1.53 confirms this directly. This is precisely the payoff profile predicted for an insurance provider, and it stands in contrast to the second strategy, whose skewness of −0.09 was essentially symmetric.

Both documented frost catastrophes produce losses. The July 2021 trade is the second-largest loss in the sample, and June 1994 also falls among the five worst. The largest single loss, August 2025, was not a frost event; it is discussed below. The strategy loses money precisely when the risk it has sold materialises.

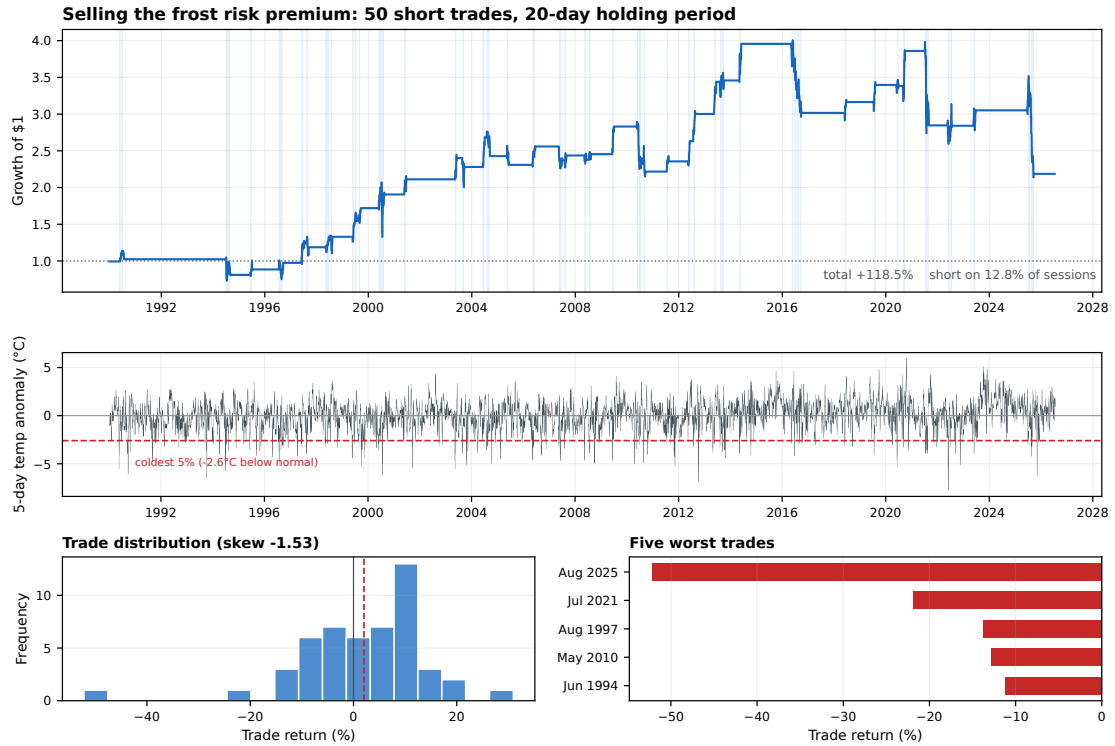


Figure 7.6: Equity curve, the five-day temperature anomaly against its fifth-percentile trigger, the distribution of trade returns, and the five worst trades. Shaded bands on the equity panel mark sessions on which the strategy is short.

The largest loss in Figure 7.6 was not driven by frost. The August 2025 trade lost 46%, despite the signal being triggered by a genuine cold anomaly. Over the holding period, however, coffee prices were dominated by the introduction of a 50% US tariff on Brazilian imports from 1 August 2025, which disrupted trade flows and contributed to certified inventories falling to multi-year lows. This highlights an important limitation of the strategy: it only responds to temperature and cannot account for other shocks affecting coffee prices. In this case, the loss was driven by tariff risk rather than the frost-risk premium the strategy is designed to capture.

7.4 Index, Wrapper and Stakeholders

the previous subsections specify the frost premium strategy as a set of trading rules. To be offered to a client it must be stated as an index: a single published series, computable from public inputs, with no discretionary input at any point. This section gives that specification.

7.4.1 The climate state C

Let

$$C_t = T_t^{MG} \quad (7.1)$$

denote the daily minimum 2m air temperature as usual, in the Sul de Minas producing region of Minas Gerais, with natural filtration $\mathcal{F}_t$.

7.4.2 The climate signal ξ

Let $\bar{c}(d)$ denote the day-of-year climatology of C_t , smoothed with a ± 7 -day window, and define the anomaly and its five-day trailing mean

$$a_t = C_t - \bar{c}(d(t)), \quad A_t = \frac{1}{5} \sum_{i=0}^4 a_{t-i}. \quad (7.2)$$

Let q_t denote the fifth percentile of $\{A_s\}_{s \leq t}$, re-estimated annually using only data available before the current season opens. The signal takes values in $\mathcal{X} = \{-1, 0\}$, with neutral signal $\xi^0 = 0$, and is defined by

$$\xi_t = \begin{cases} -1 & \text{if } A_t < q_t, A_{t-1} \geq q_t, t \in \mathcal{W}, \text{ and } \xi_s = 0 \forall s \in (t-20, t), \\ -1 & \text{if } \xi_{t-1} = -1 \text{ and fewer than 20 trading days have elapsed since entry,} \\ 0 & \text{otherwise,} \end{cases} \quad (7.3)$$

where $\mathcal{W} = \{t : 1 \text{ May} \leq t \leq 1 \text{ September}\}$ is the season window. The first line is the entry condition: a fresh crossing into the coldest five per cent of the anomaly distribution, inside the frost season, with no position currently open. The third condition on the first line is the non-overlap rule, so that at most one position is open at any time; signals arriving while $\xi_{t-1} = -1$ are recorded but not acted upon. The second line holds the signal for the twenty-day horizon. A position entered before 1 September runs to its full twenty-day exit even if that exit falls outside $\mathcal{W}$: the window gates entries, not exits. Note that ξ_t is $\mathcal{F}_t$ -measurable, and by (4.23) the rule below uses ξ_{t-} , so no position is formed on information not yet available.

7.4.3 The positioning rule g

The traded asset is the ICE Coffee “C” (KC) front-month futures contract, so $S_t = (B_t, S_t^{KC})$ in the notation of (4.22). The rule is

$$\theta_t = g(t, \xi_{t-}) = (\theta_t^0, \xi_{t-}), \quad (7.4)$$

that is, the holding in KC equals the signal itself: -1 contract per unit of index notional when short, zero otherwise, with θ_t^0 the residual cash holding in the money-market account. Since the neutral allocation is flat, $g(t, \xi^0) = (\theta_t^0, 0)$, and by (4.26) the climate position is

$$\Pi_t^C = \theta_t - g(t, \xi^0) = \xi_{t-}, \quad (7.5)$$

So the climate position consists entirely of the KC holding, which equals the signal ξ_{t-} . The contract held is always the nearest unexpired KC delivery month. The index rolls to the next contract in the KC cycle (March, May, July, September, December). If $\theta_t \neq 0$ on a roll date, the position is closed in the expiring contract and reopened in the new front month at that day’s settlement, with returns computed on a ratio back-adjusted basis so the roll introduces no artificial jump into I_t . A trade spanning a roll therefore pays transaction cost twice. Transaction costs are charged at $c = 5$ basis points per side.

Writing r_t^{KC} for the daily return of the held (rolled, back-adjusted) front-month contract, the index return and level are

$$r_t^I = \theta_{t-1} r_t^{KC} - c |\theta_t - \theta_{t-1}|, \quad I_t = I_{t-1}(1 + r_t^I), \quad I_0 = 100. \quad (7.6)$$

Every term on the right is determined by public temperature and settlement data as of each close, so I_t is fully reconstructible by a third party and admits no discretionary override.

7.4.4 The wrapper

The index is not directly investable by most end clients, since holding it requires an ICE futures account, daily margining, and monitoring of the signal and roll calendar. It is therefore delivered as a total return swap on I_t written by a bank’s commodity structuring desk, the sponsor (lets say MUFG).

The reference obligation is I_t as defined above, published daily by the sponsor or an independent calculation agent from the same public inputs. At each quarterly reset the investor receives $I_t/I_{t_0} - 1$ on the swap notional and pays a running fee of 75 basis points per annum, accrued daily and netted at reset. Because the index is short-only and flat on roughly 87% of sessions (Table 7.7), there is no financing leg of the kind found in a fully funded equity total return swap; the fee compensates the sponsor for replication and operational risk, not for funding a long position.

To hedge the swap, the dealer mirrors the position taken by the index using front-month KC coffee futures. When the index is short coffee, the dealer takes an equivalent short

position in KC futures; when the index has no position, the dealer also remains flat. The futures hedge is rolled on the same dates as the index. Since the swap payoff is linked to the performance of the index, gains and losses on the futures position should largely offset the amounts the dealer must pay or receive under the swap. The dealer is therefore not deliberately taking a view on coffee prices or frost risk. Its main remaining exposure comes from execution: the price at which it can actually trade may differ from the official settlement price used by the index, particularly when rolling or trading a large position. The product fee is intended to compensate the dealer for these risks, including slippage, timing differences and liquidity costs

Given the -46% single-trade loss realised in August 2025, the swap should carry an early-termination trigger if the investor's mark-to-market loss exceeds a pre-agreed fraction of notional, of the order of 25% , within a single twenty-day window. This forces a re-margining conversation rather than allowing a loss of that magnitude to accumulate before either party acts.

7.4.5 Investor, sponsor and capacity

The natural buyer is a specialist insurance-linked securities or reinsurance-linked fund rather than a directional commodity investor. The index is structurally a premium-selling, negatively skewed carry exposure, with realised skewness of -1.53 and a hit rate of 60% (Table 7.8), which is the same shape as catastrophe bond or parametric weather risk. Such a fund already holds the capital, the pricing discipline and the mandate for infrequent-payout premia, and Brazilian frost is genuinely diversifying against a book of hurricane, earthquake or mortality risk.

The natural seller is a bank like MUFG with an established ICE soft commodities franchise and an existing coffee clearing relationship, since it already operates the hedging and delivery infrastructure for producer and roaster clients. The marginal cost of adding a systematic overlay to an existing coffee book is small relative to establishing the capability from nothing.

Chapter 8

Conclusion

To conclude, This dissertation investigated whether climate-related information can be converted into a financially meaningful position in agricultural commodity markets. The original objective was to design a climate derivative that would allow investors or commercial users to hedge, or express views on, climate-related risks. However, the analysis showed that a bespoke climate derivative is difficult to implement in practice.

The main contribution of this dissertation is therefore twofold. First, it shows that climate-related information can be quantified and linked to coffee futures returns using both climate news indices and physical weather variables. Second, it shows that a QIS can provide a more practical implementation of climate positioning than a bespoke derivative, while still preserving a clear connection to the underlying climate mechanism. The final strategy demonstrates that climate signals can be translated into systematic trading rules, but also that their risk, limitations and commercial viability must be assessed carefully.

Future research could improve the analysis in several directions. The first is to refine the weather signal. This dissertation used reanalysis temperature data for key Brazilian coffee-growing regions, but frost is highly localised and also depends on elevation, humidity, wind speed, cloud cover and local topography. A more detailed signal could combine station-level observations, satellite data, soil moisture and agronomic information on the location of coffee farms. This would help separate harmless cold anomalies from genuinely damaging frost events.

A second extension would be to use weather forecasts rather than realised temperatures. A live strategy should react to information available before or during the repricing process, rather than only after the minimum temperature has been observed. Forecast-based signals could test whether the market overprices or underprices frost risk before the event occurs, and whether the premium decays as forecasts are revised.

A third direction would be to extend the strategy across the futures curve. Severe frost can affect both near-term supply and future production capacity, so different maturities may respond differently to the same shock. Future work could therefore study calendar spreads, curve slope and roll yield around frost events, rather than trading only the front-month

futures contract.

Finally, future research should focus on risk management and testing we have not considered. Since the frost premium strategy has insurance-like downside risk, position sizing should account for margin requirements, liquidity and stress losses during severe frost episodes. Volatility targeting, drawdown-based exposure limits or stop-loss rules could be tested. The same framework could also be applied to other agricultural commodities such as Robusta coffee, cocoa, sugar, wheat or corn. This would help determine whether the result found here is specific to Brazilian Arabica coffee, or whether it is part of a broader climate-risk premium in agricultural futures markets.

Declaration of generative AI and AI-assisted technologies in the preparation process

Generative AI tools (ChatGPT and Claude opus/fable 5) were used during the preparation of this dissertation to support coding, the creation and refinement of figures and proofreading. I also developed an AI-based agent to provide indicative marking and feedback on draft versions of the thesis. All research decisions, analysis, interpretation of results, and final written content were reviewed and determined by me. I remain fully responsible for the work presented in this dissertation.

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